THEORY

Training Program 01

Advanced NLP & Large Language Models (LLMs)

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- **M01:** Text Preprocessing
- **M02:** Word and Sentence Embeddings
- **M03:** Transformer Architectures
- **M04: Entity-Centric NLP**
- **M05:** Applied NLP Tasks
- **M06:** Multimodal NLP
- **M07:** LLM Fundamentals
- **M08:** Prompt Engineering
- **M09:** Fine-Tuning & PEFT
- **M10:** RAG & Real-World Projects

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MODULE M01

TEXT PREPROCESSING

CHAPTER M04

Entity-Centric NLP

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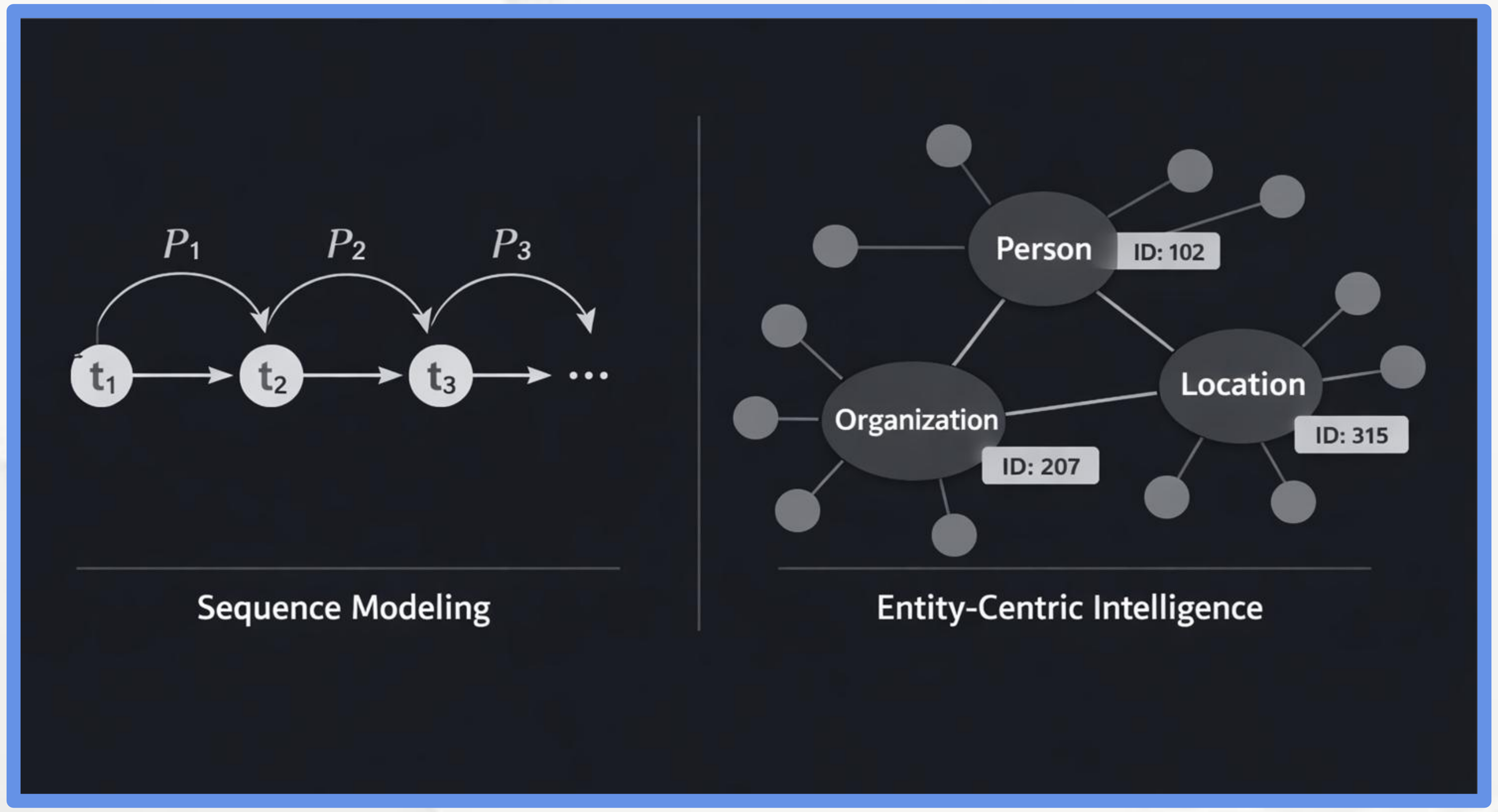
Chapter 4.1 Introduction to Entity-Centric NLP

From Sequence Modeling to Entity-Centric Intelligence

- Traditional NLP models language as token sequences and probabilities.
- High-reliability AI requires stable, identifiable real-world objects.
- Entity-Centric NLP shifts from linguistic tokens to knowledge objects.
- Entities enable tracking, memory, and reasoning across documents.
- This marks the transition from fluent language to grounded intelligence.



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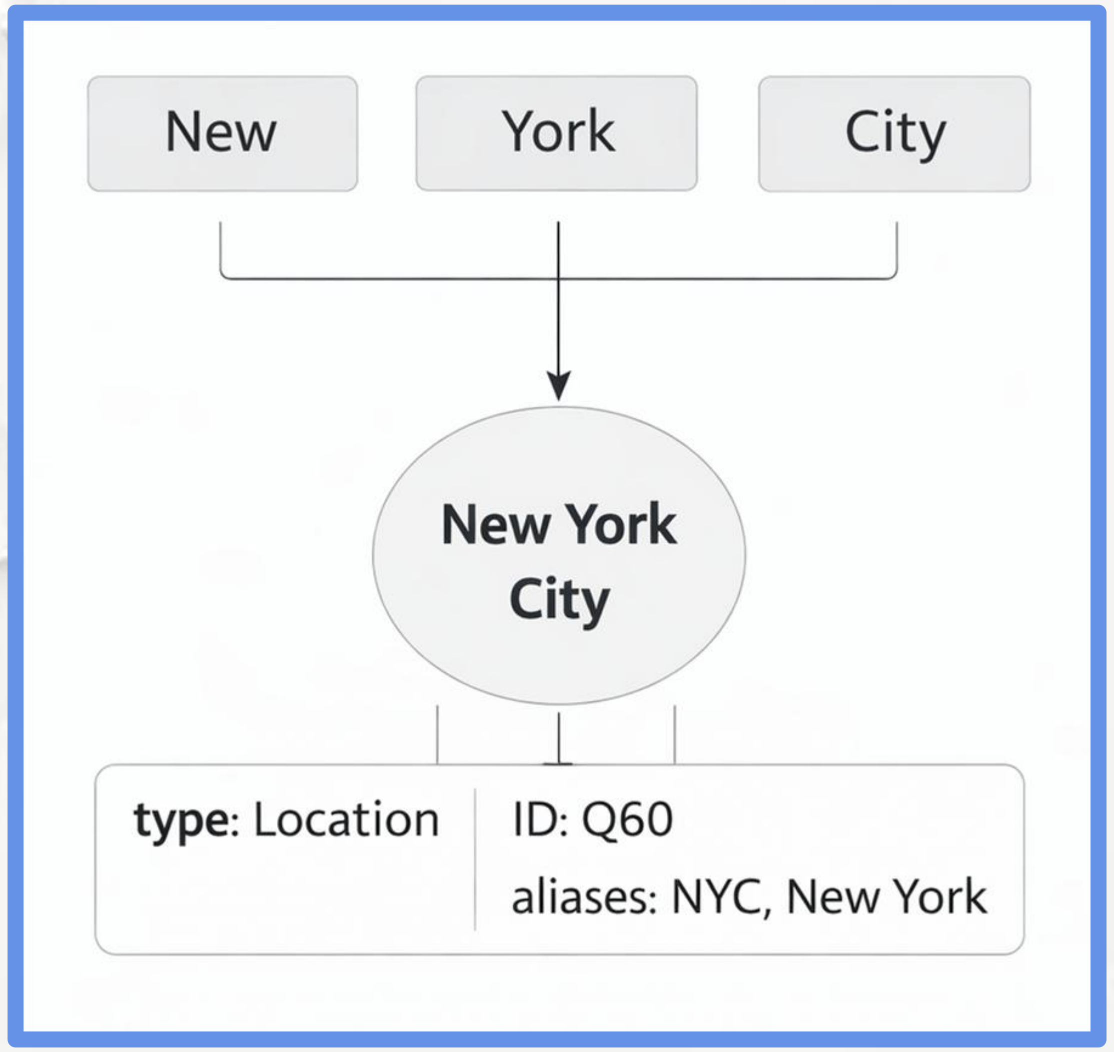
Chapter 4.1 Introduction to Entity-Centric NLP

What Is an Entity? The Atom of Knowledge

- An entity is a unique, persistent real-world or domain-specific object.
- Unlike tokens, entities maintain identity across contexts and documents.
- Examples include people, locations, organizations, laws, or chemicals.
- Entities are the stable units stored in knowledge graphs and databases.
- They enable long-term memory and cross-document reasoning.



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Chapter 4.2 NER: Identification and Classification (Index)

What Is Named Entity Recognition?

- NER is a sequence tagging task over tokenized text.
- Each token is assigned a label indicating entity membership.
- Labels encode both boundaries and semantic categories.
- The output transforms raw text into structured entity spans.
- NER provides the foundation for all downstream entity tasks.

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Chapter 4.2 NER: Identification and Classification (Index)

Standard Entity Types and Taxonomies

- NER systems classify entities into predefined semantic categories.
- Core universal types form the base of most industrial systems.
- Extended taxonomies adapt NER to specific domains.
- Correct typing is critical for downstream reasoning and linking.

Core entity categories

- PER – People and individuals.
- ORG – Organizations and institutions.
- LOC – Locations (non-political).
- GPE – Countries, cities, geopolitical regions.

Extended / domain types

- PER – People and individuals.
- ORG – Organizations and institutions.
- LOC – Locations (non-political).
- GPE – Countries, cities, geopolitical regions.

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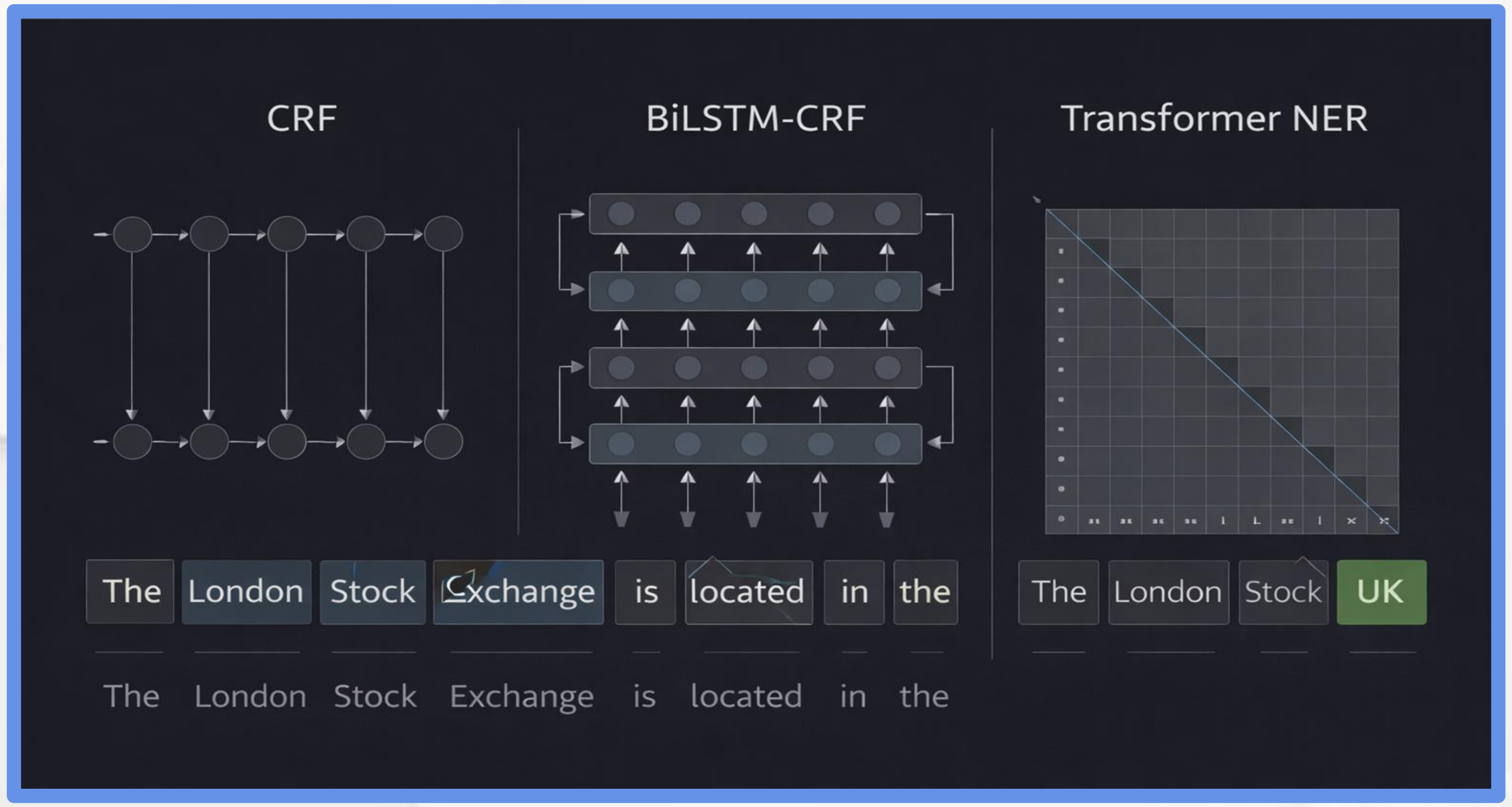
Chapter 4.2 NER: Identification and Classification (Index)

Sequence Tagging Approaches

- NER is framed as a sequence labeling problem.
- Models predict a label for each token in context.
- Approaches evolved from probabilistic to deep neural architectures



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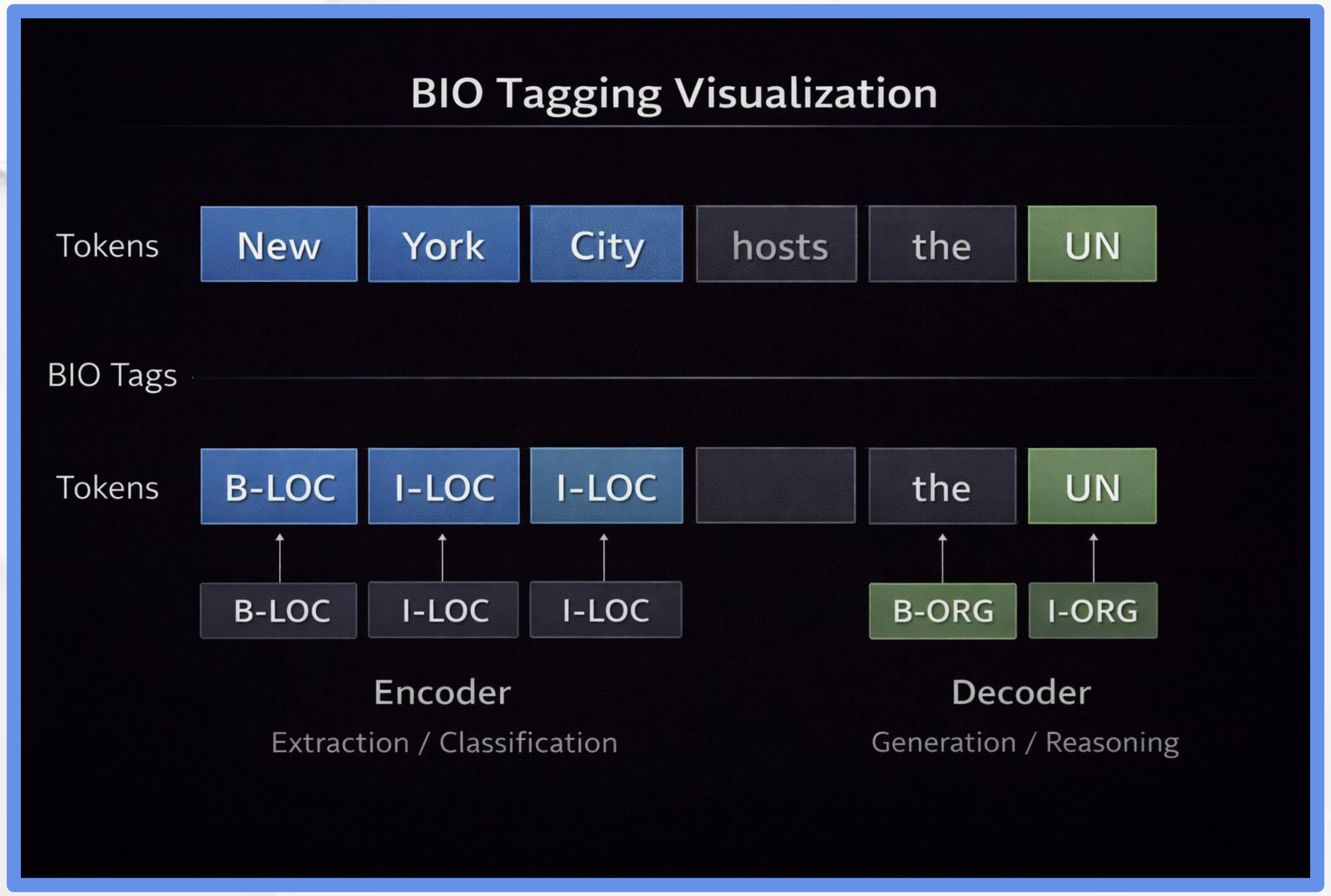
Chapter 4.2 NER: Identification and Classification (Index)

Annotation Schemes: BIO / IOB Tagging

- Multi-token entities require boundary-aware labeling.
- BIO (or IOB) is the most widely used annotation scheme.
- Each token receives a boundary + type labe.



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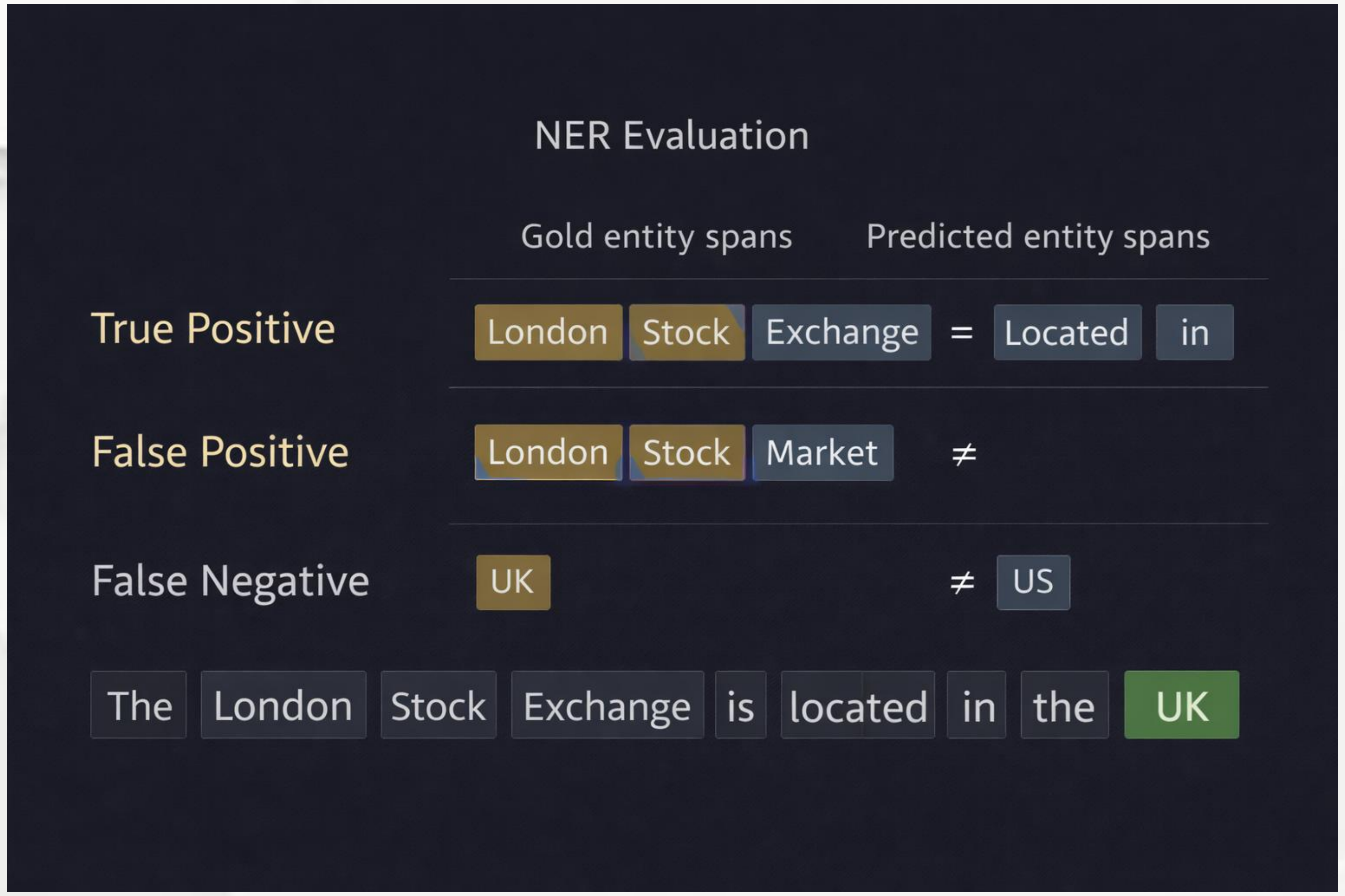
Chapter 4.2 NER: Identification and Classification (Index)

Evaluation Metrics for NER

- NER evaluation is span-based, not token-based.
- Correct prediction requires exact boundary + label match.
- Partial matches count as errors.



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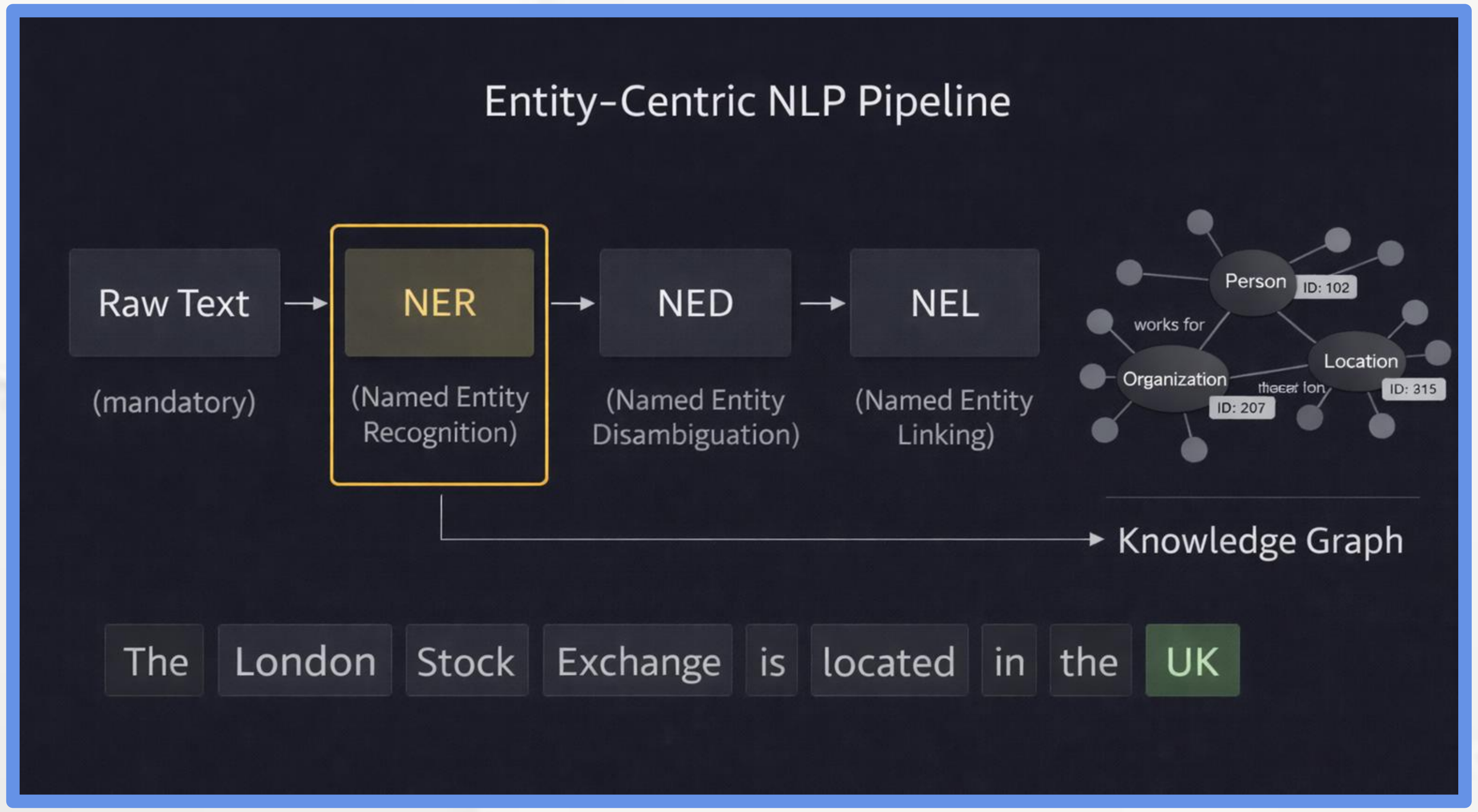
Chapter 4.2 NER: Identification and Classification (Index)

Why NER Is the Foundation of Entity-Centric NLP

- NER is the entry point to entity intelligence.
- All downstream tasks depend on accurate detection.
- Errors propagate if NER fails.
- NER → NED → NEL → Relation Extraction → Knowledge Graphs



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4.1 Introduction to Entity-Centric NLP

4.2 Named Entity Recognition (NER): Identification and Classification

4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

4.5 Advanced Domain-Specific Entity Extraction

4.6 Summary and The Entity Lifecycle

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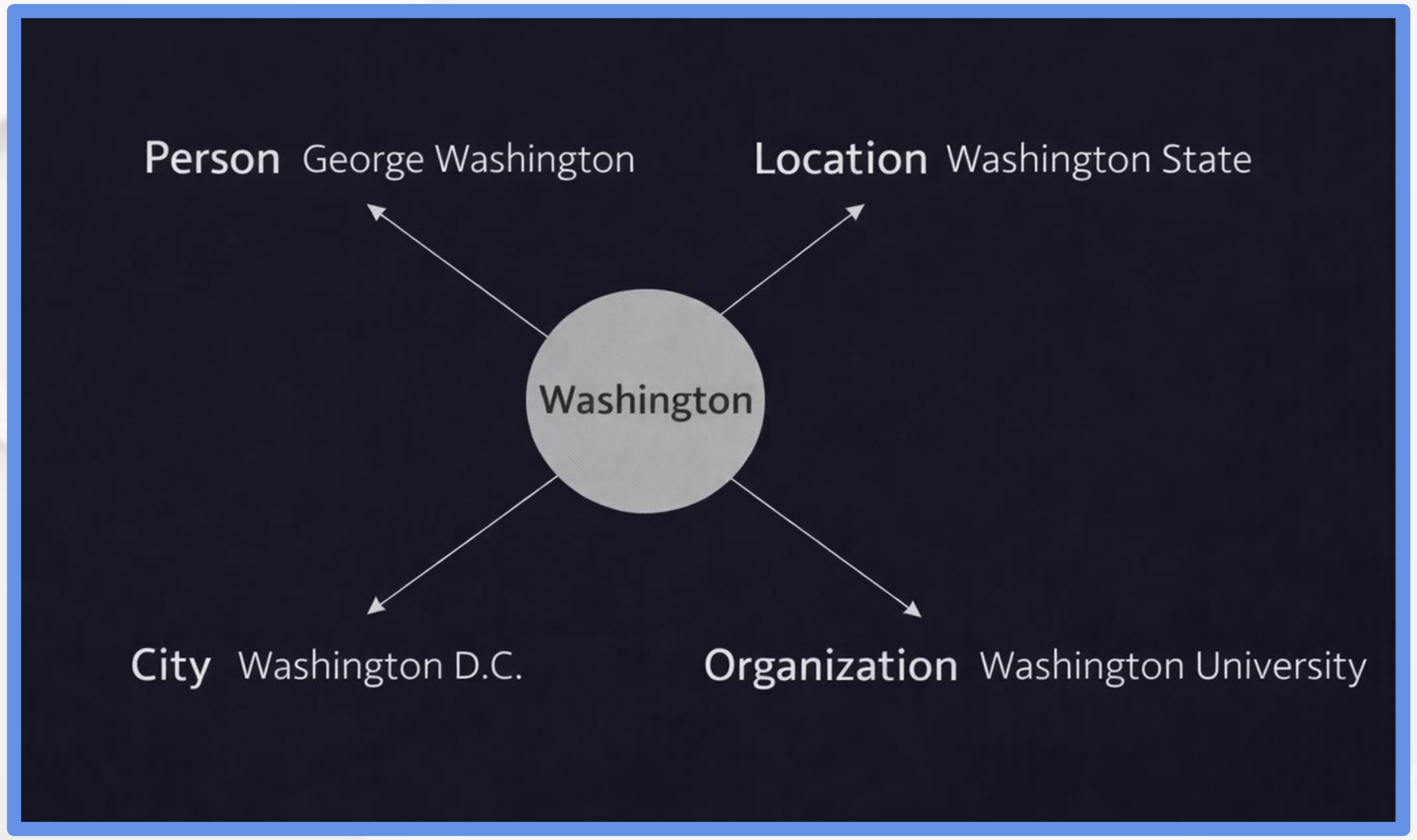
Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

The Problem of Entity Ambiguity

- The same surface name can refer to multiple real entities.
- Ambiguity is unavoidable in natural language.
- NER alone cannot determine the correct referent.
- Unresolved ambiguity leads to factual errors.



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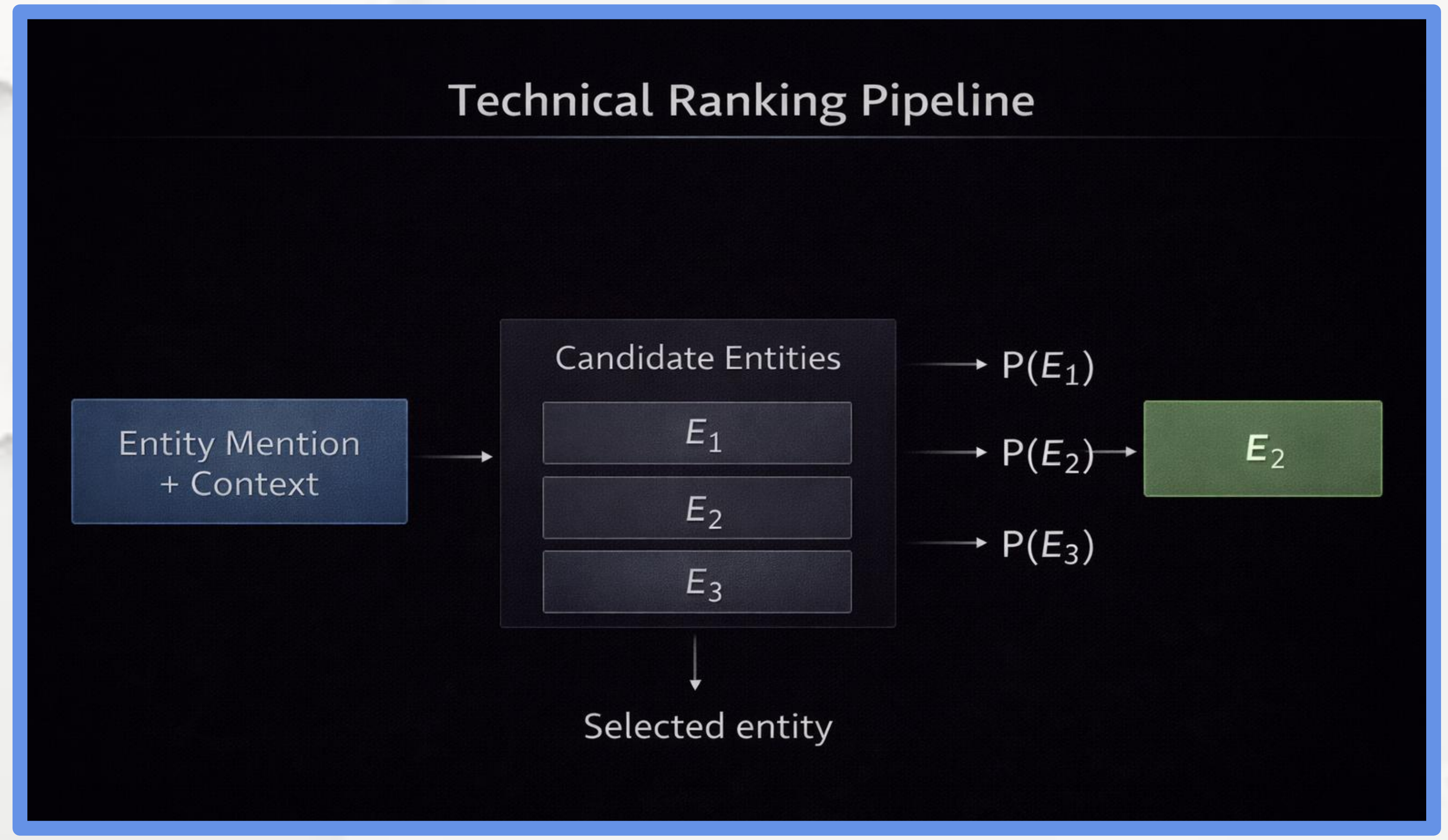
Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

What Is Named Entity Disambiguation (NED)

- NED selects the correct entity among candidates.
- It is modeled as a ranking problem.
- Input: mention + context.
- Output: single most probable real-world entity.



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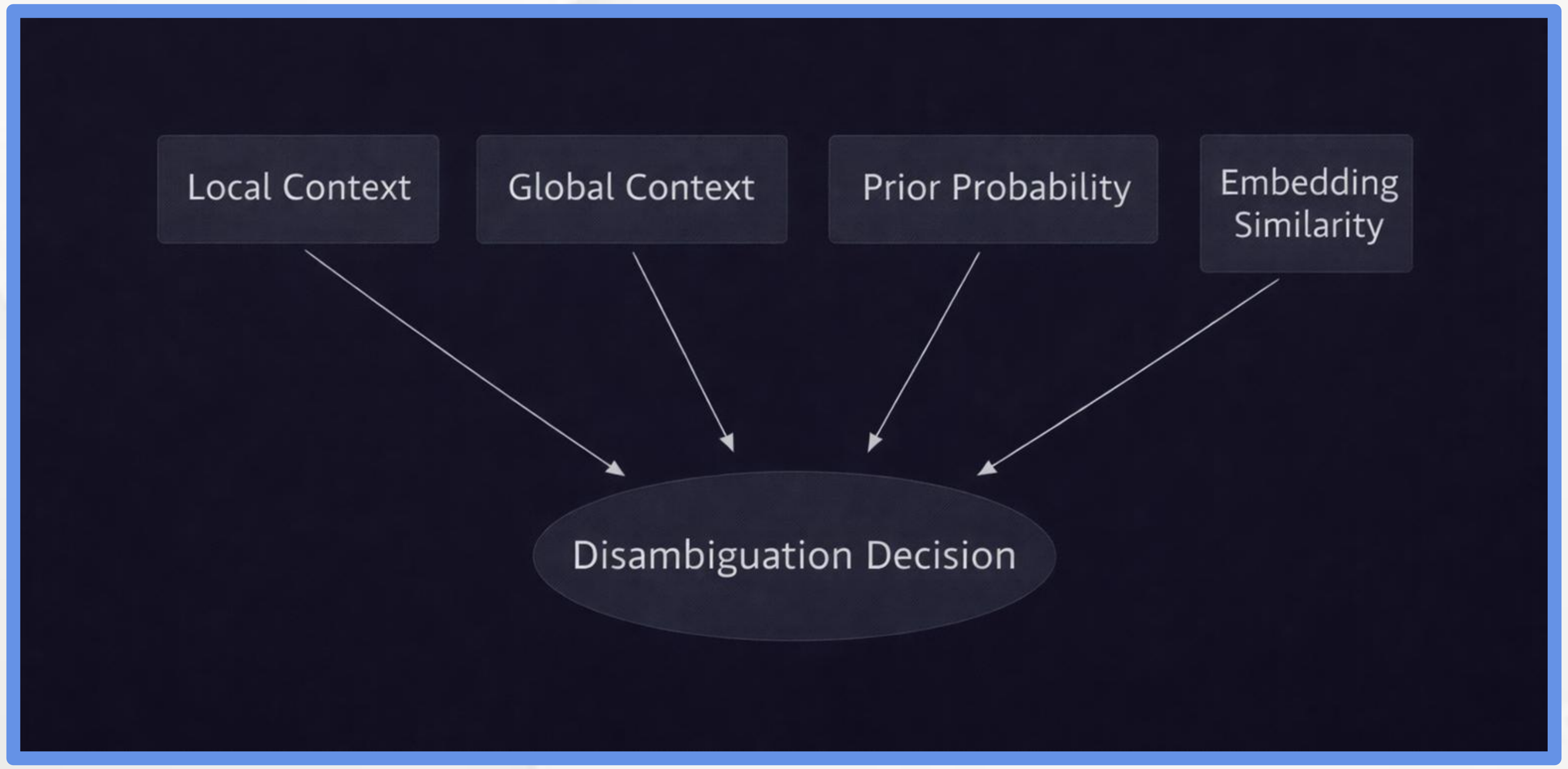
Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

Signals Used in Entity Disambiguation

- Local context around the mention.
- Global document coherence.
- Prior probability from knowledge bases.
- Semantic similarity in embedding space.



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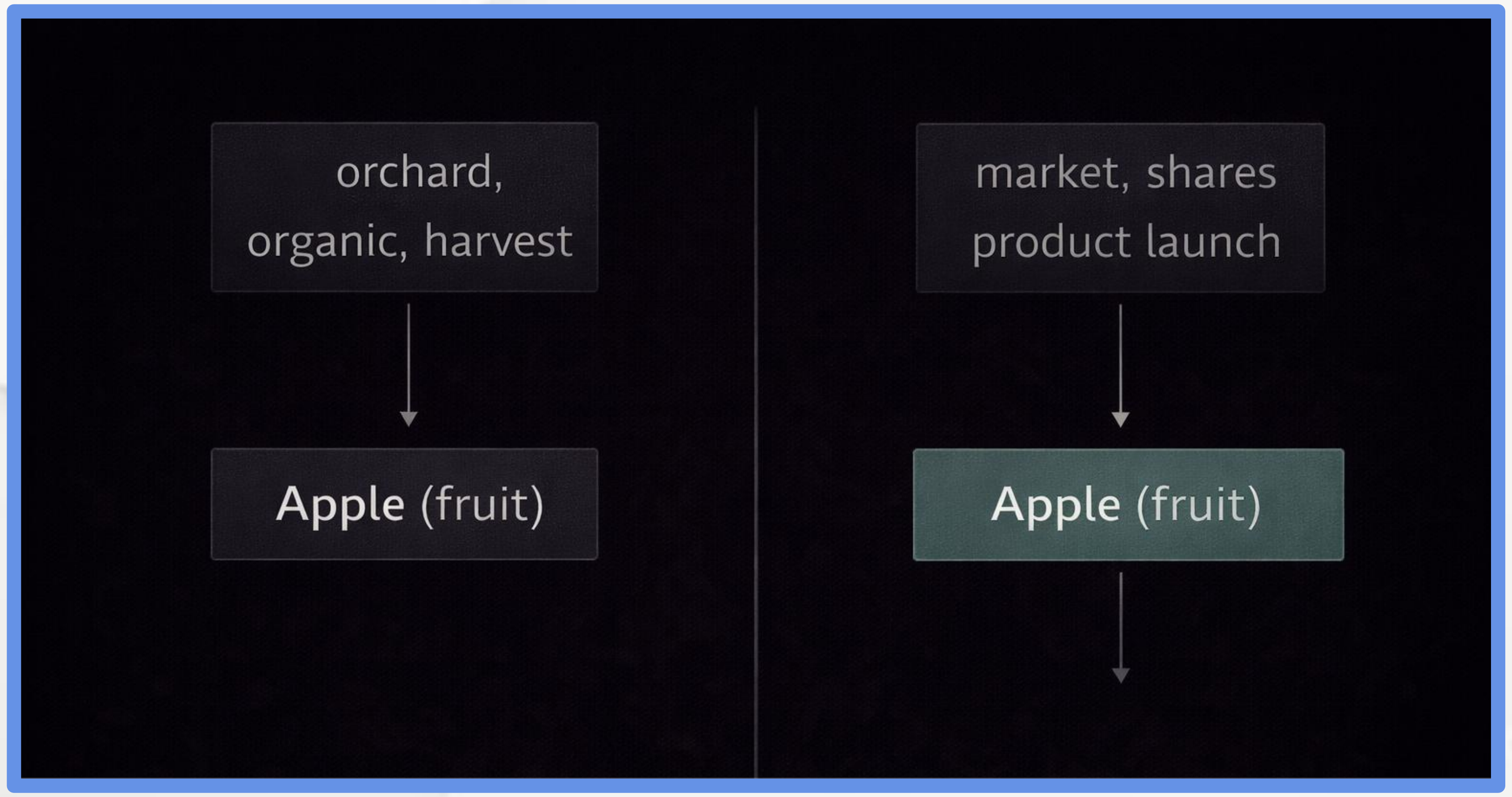
Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

Example: Resolving Ambiguous Mentions

- Context determines the entity meaning.
- Semantic cues guide the resolution.
- Same word, different real-world entities.
- Correct resolution enables safe reasoning.



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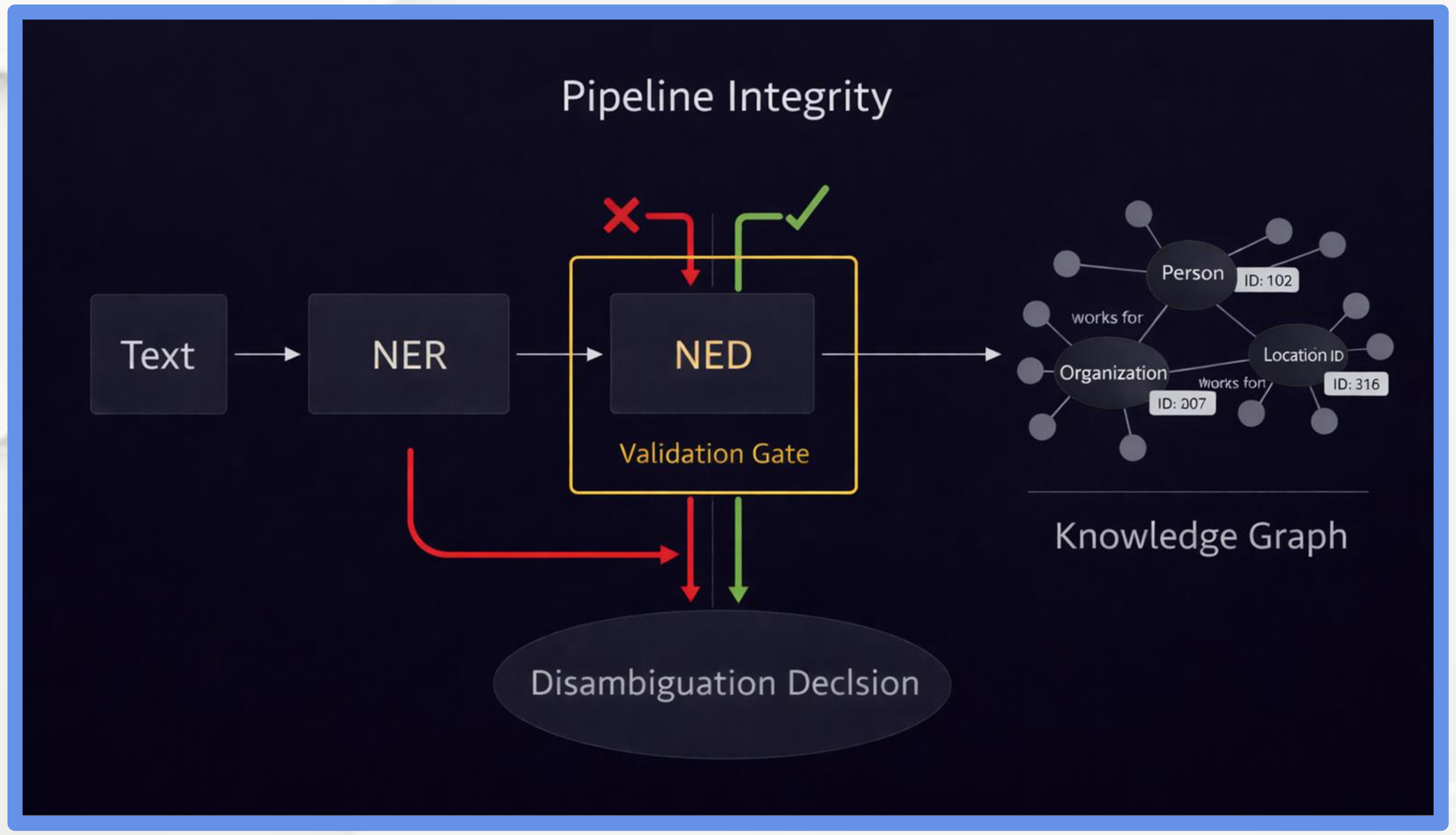
Chapter 4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

Why NED Is Critical for Knowledge Systems

- Prevents incorrect facts entering knowledge graphs.
- Enables precise analytics and reasoning.
- Required before entity linking (NEL).
- Core reliability layer in enterprise AI.



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4.1 **Introduction to Entity-Centric NLP**

4.2 Named Entity Recognition (NER): Identification and Classification

4.3 Named Entity Disambiguation (NED): Resolving Entity Ambiguity

4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

4.5 Advanced Domain-Specific Entity Extraction

4.6 Summary and The Entity Lifecycle

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Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

What Is Named Entity Linking (NEL)

- NEL connects text mentions to real-world entities.
- Each entity is resolved to a unique identifier.
- Eliminates ambiguity across aliases and synonyms.
- Enables persistent, machine-readable references.

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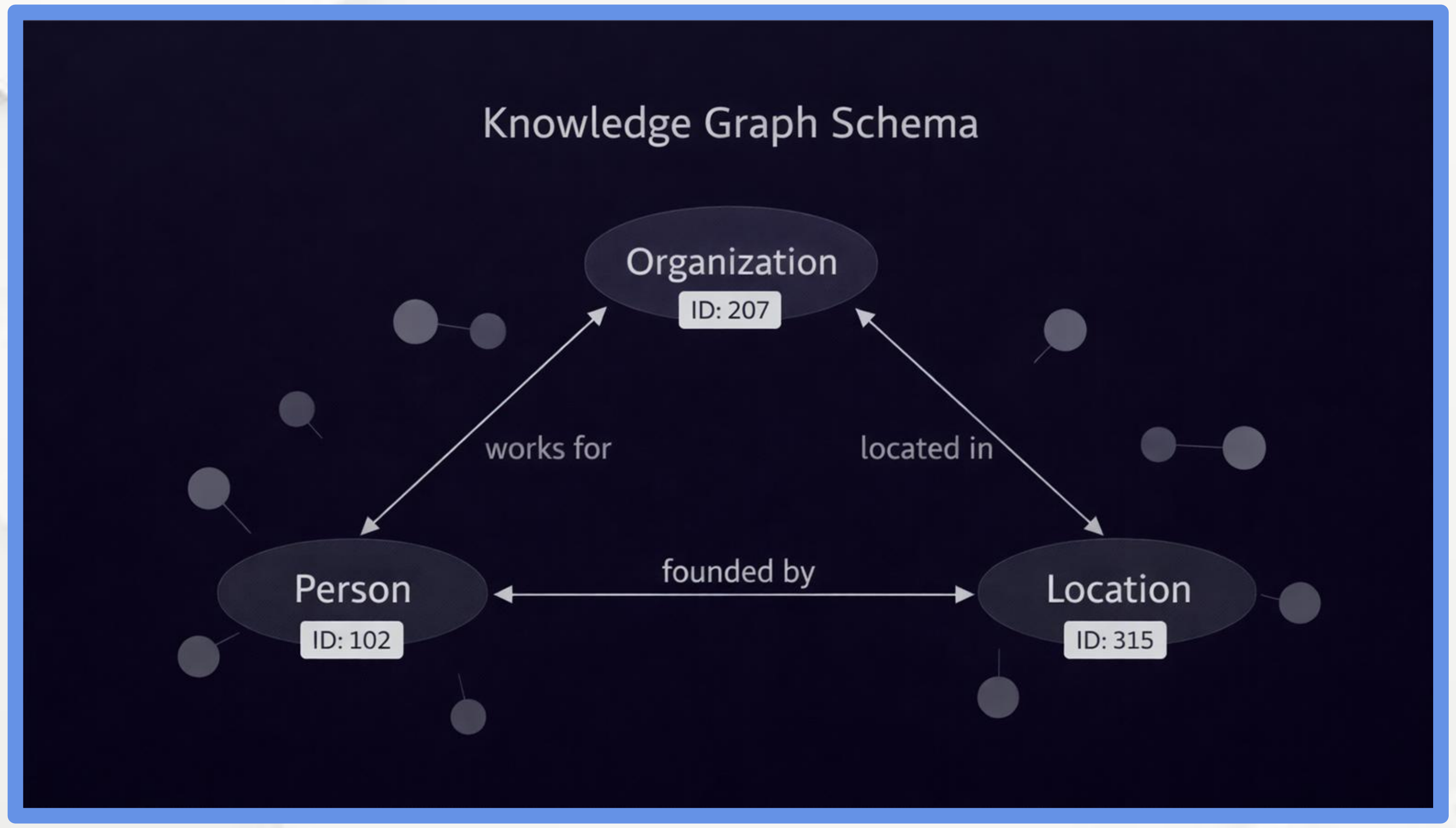
Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

Knowledge Bases and Unique Identifiers

- Knowledge Bases store structured world facts.
- Entities are identified by global IDs.
- Relations are stored as subject–predicate–object triples.
- Enables reasoning beyond raw text.



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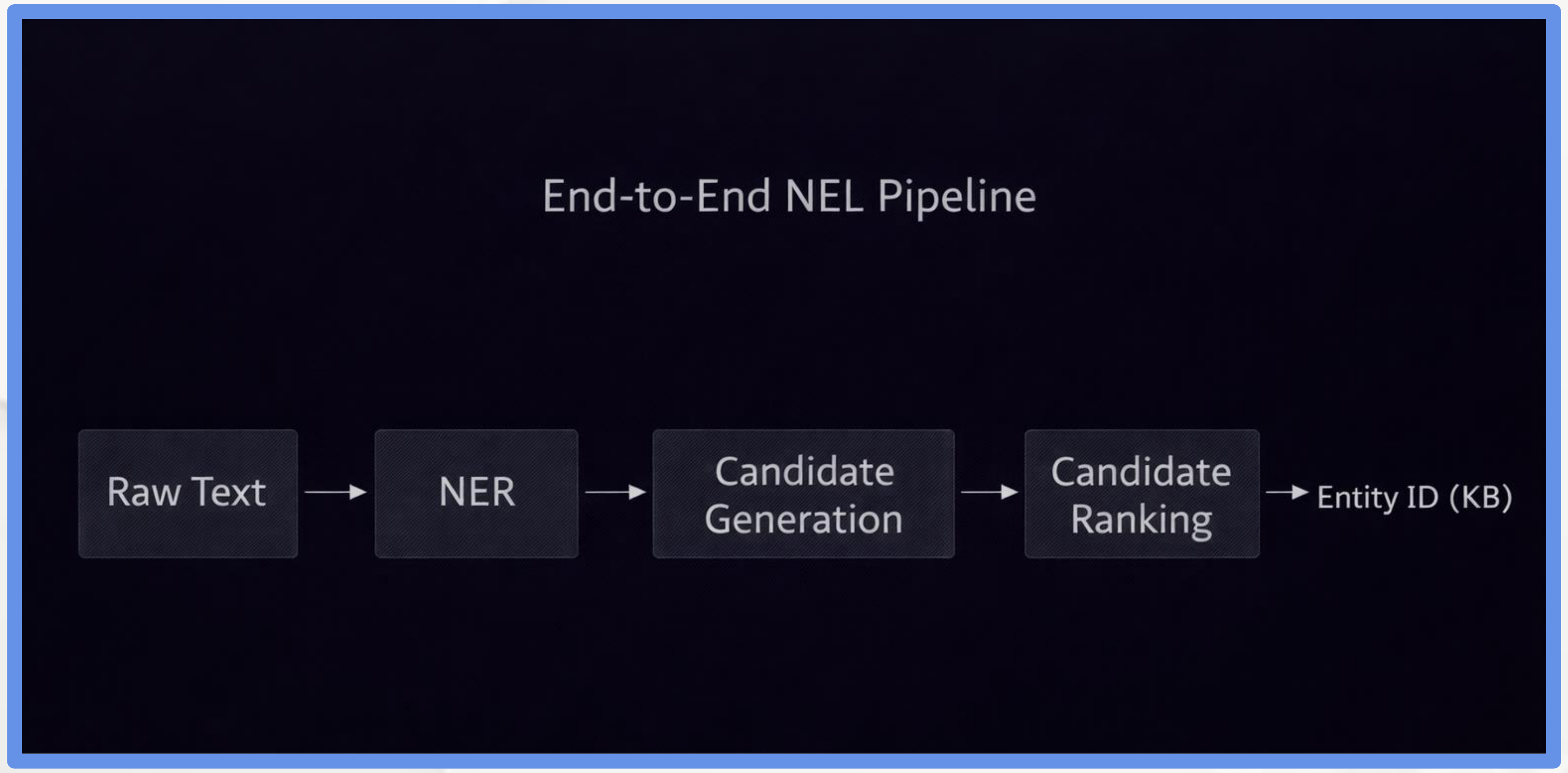
Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

The NEL Engineering Pipeline

- Detect entity mentions (NER).
- Generate candidate entities.
- Rank candidates using context (NED).
- Link to knowledge base ID.

Handle unknown entities (NIL).



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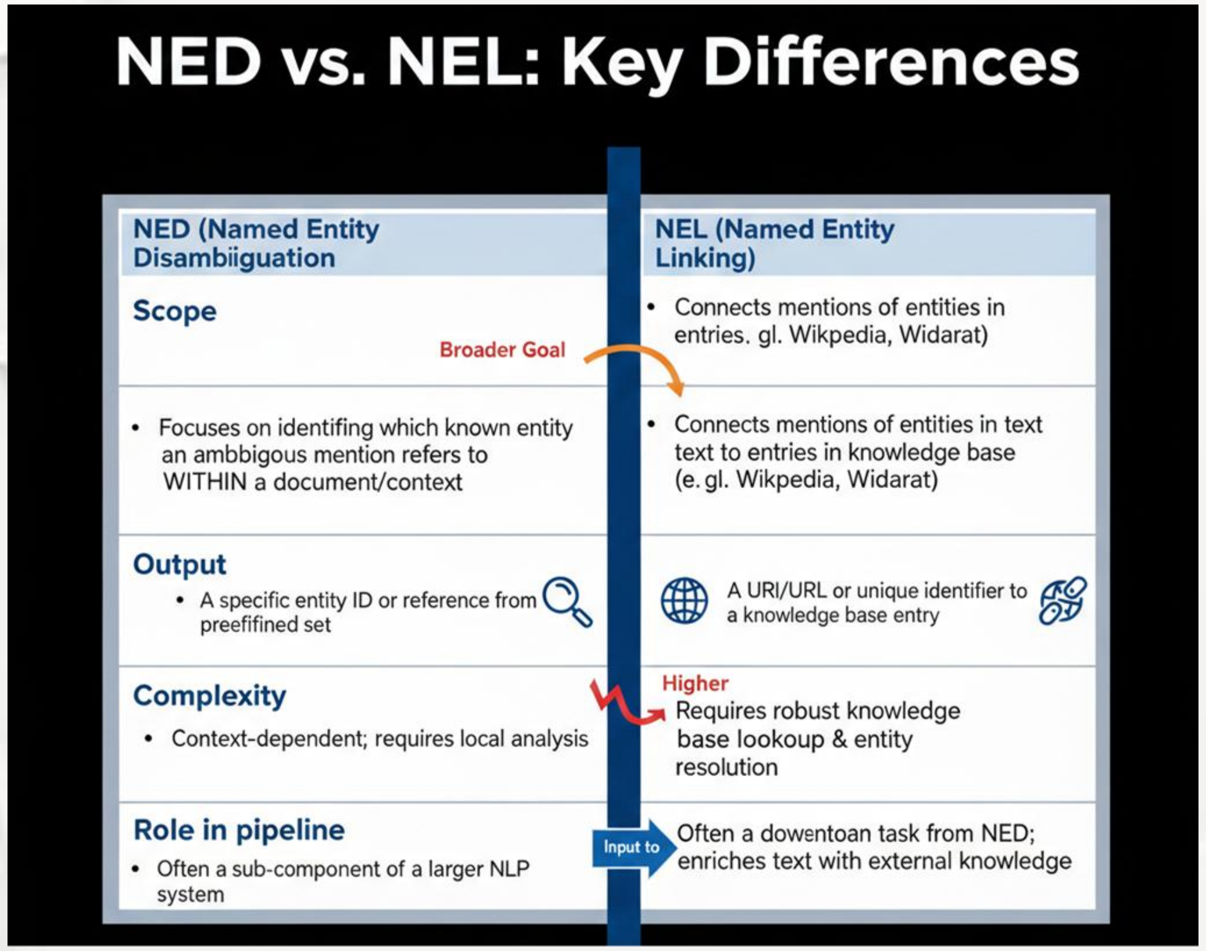
Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

NED vs NEL: Key Differences

- NED selects the correct candidate.
- NEL implements the full linking system.
- NED is a decision task.
- NEL is an end-to-end architecture.



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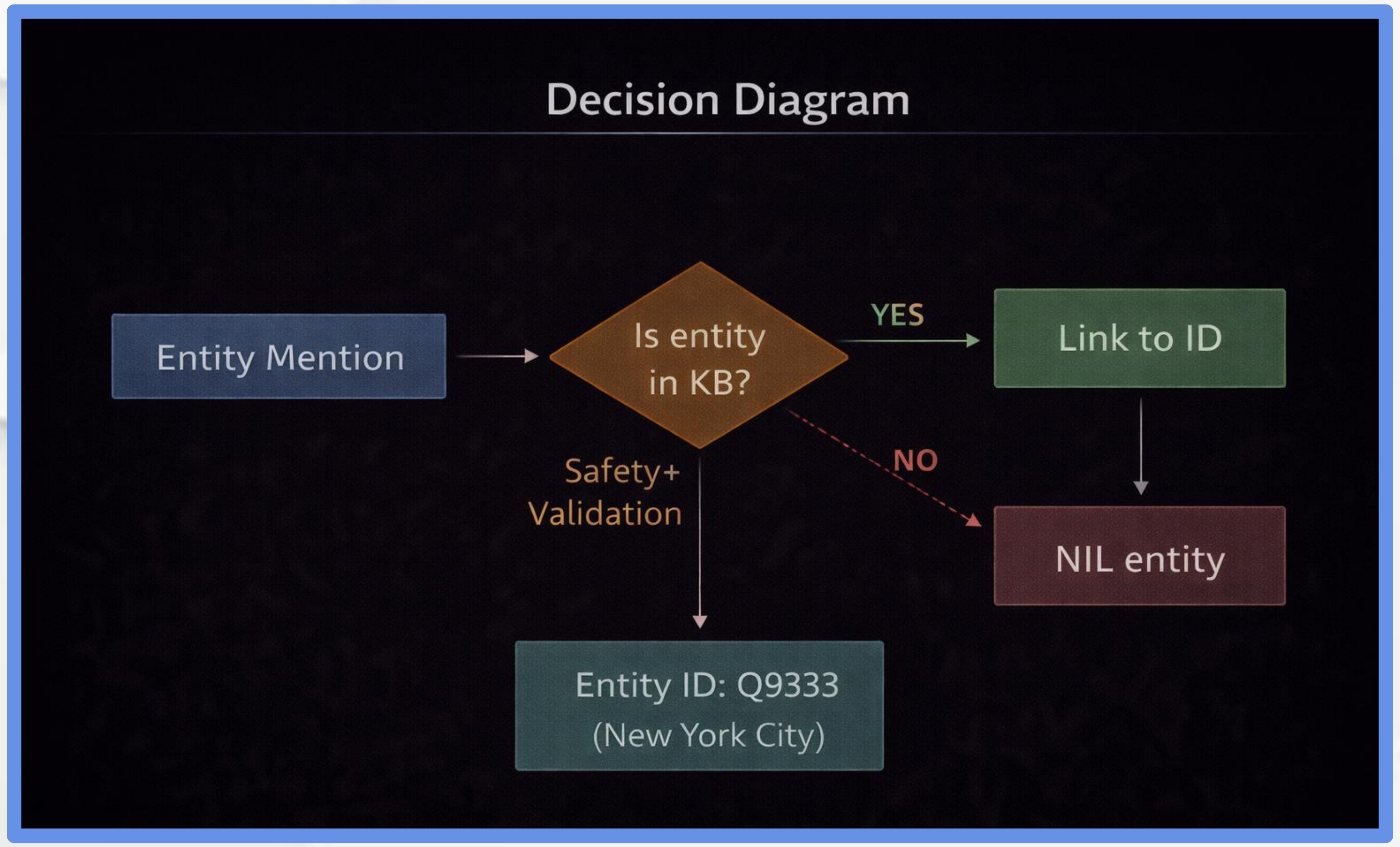
Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

NIL Detection and Real-World Constraints

- Not all entities exist in knowledge bases.
- Systems must detect NIL cases safely.
- Prevents hallucinated or false entities.
- Critical for private or emerging entities.



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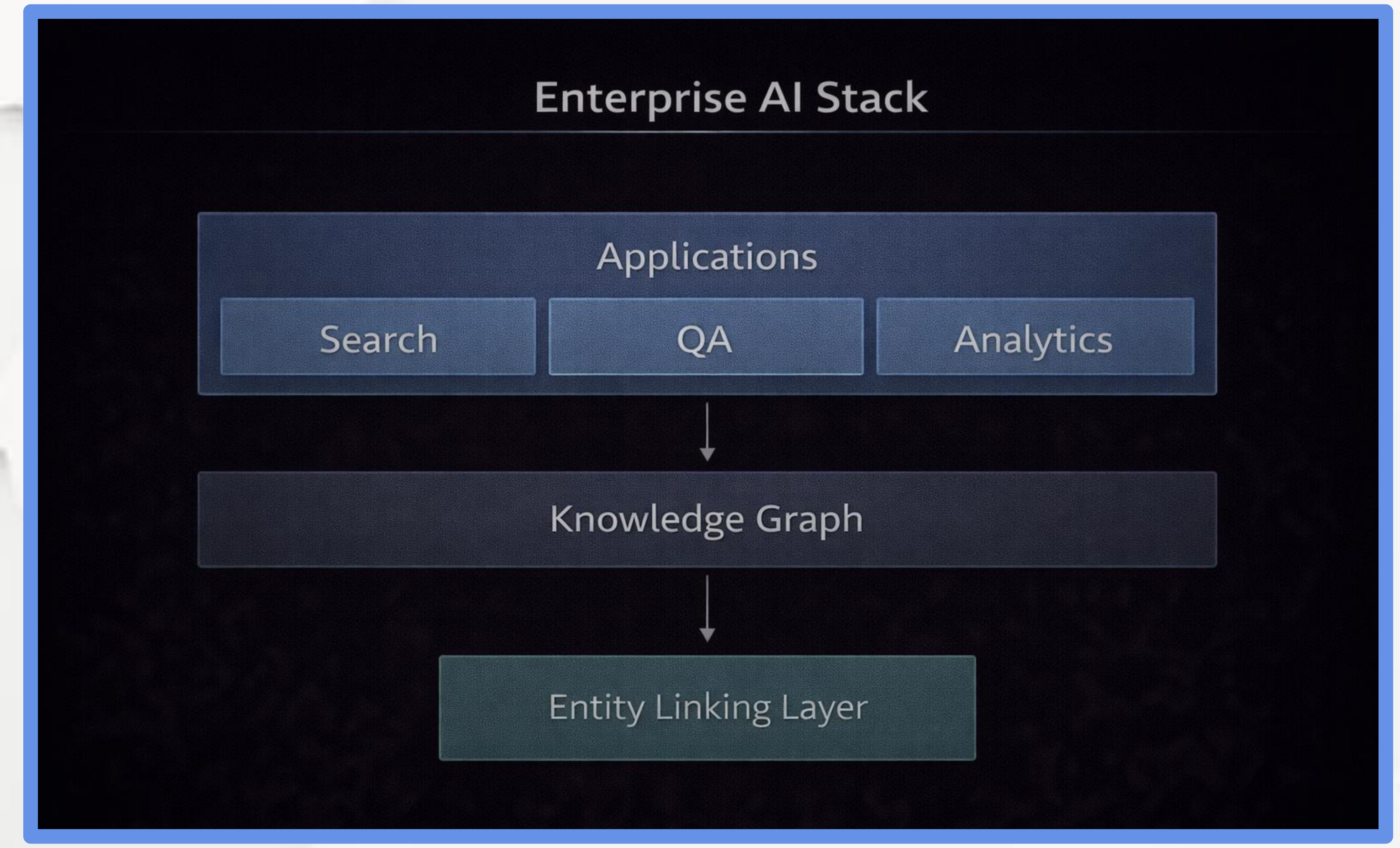
Chapter 4.4 Named Entity Linking (NEL): Mapping to Knowledge Bases

Why NEL Is Essential for Reliable AI

- Grounds language in factual reality.
- Enables explainable reasoning.
- Supports analytics and traceability.
- Foundation for enterprise knowledge graphs.



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Chapter 4.5 Advanced Domain-Specific Entity Extraction

Why Domain-Specific Entities Matter

- General NER fails in specialized domains.
- Domain entities are longer and more complex.
- Labels are highly specific and non-standard.
- Precision is critical in regulated environments.

General NER	Domain NER
• Person • Location • Organization	• Legal Clause • Chemical Compound • Medical Condition
• Person • Location • Organization	• Legal Clause • Chemical Compound • Medical Condition

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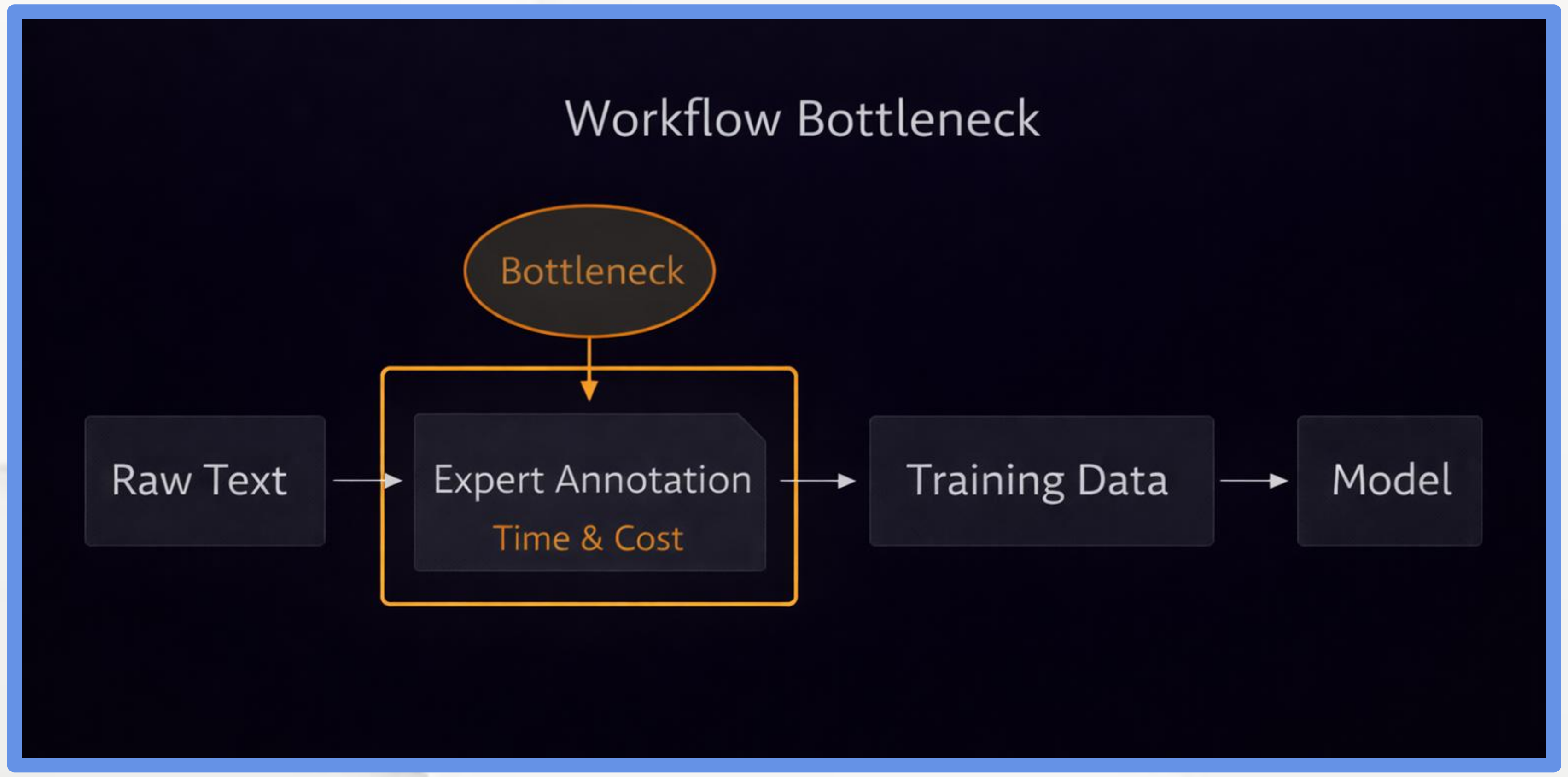
Chapter 4.5 Advanced Domain-Specific Entity Extraction

The Annotation Bottleneck

- Domain annotation requires expert knowledge.
- Manual labeling is slow and expensive.
- Data scarcity limits supervised training.
- Annotation quality impacts downstream accuracy.



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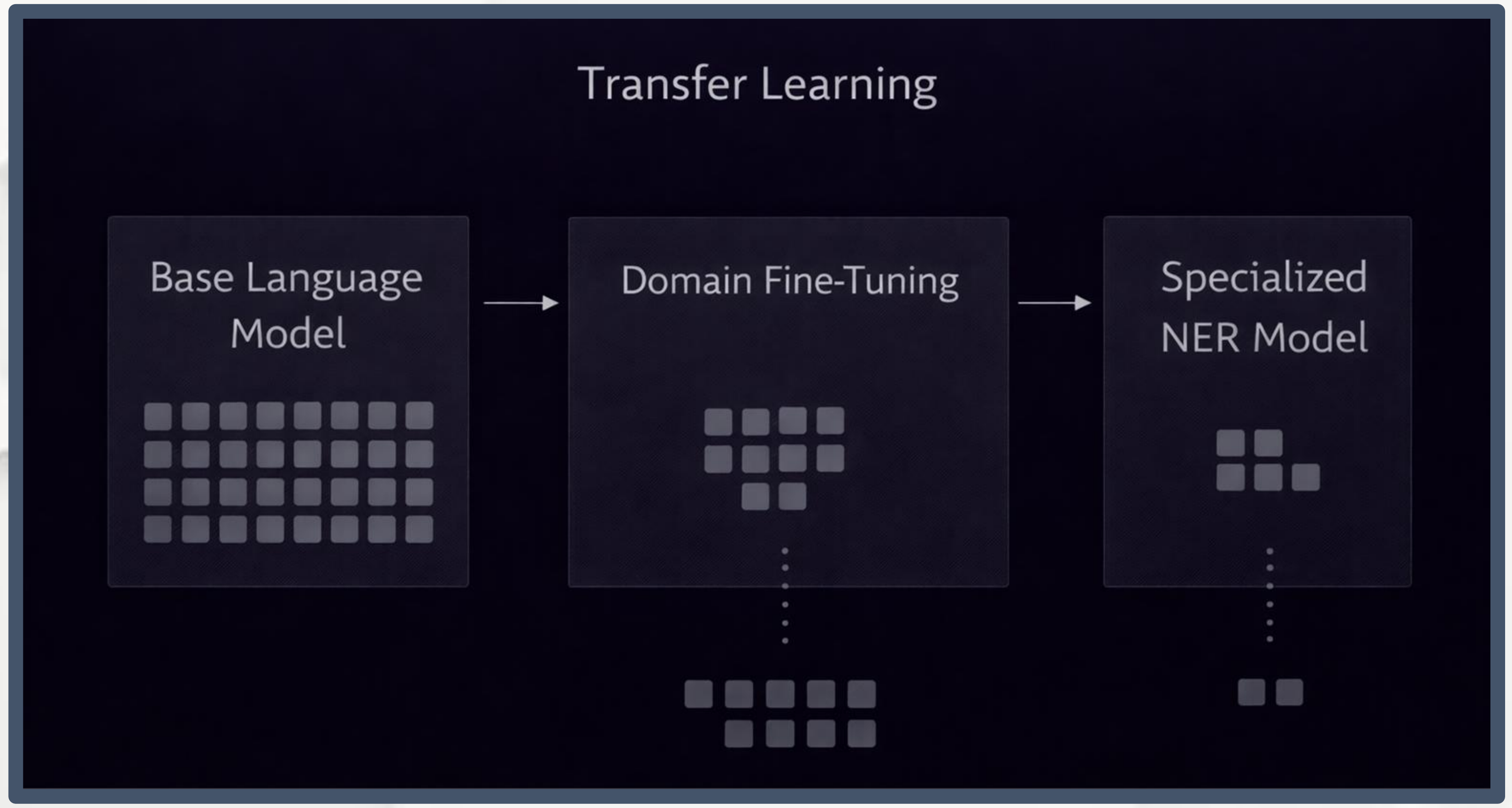
Chapter 4.5 Advanced Domain-Specific Entity Extraction

Transfer Learning for Specialized NER

- Domain annotation requires expert knowledge.
- Manual labeling is slow and expensive.
- Data scarcity limits supervised training.
- Annotation quality impacts downstream accuracy.



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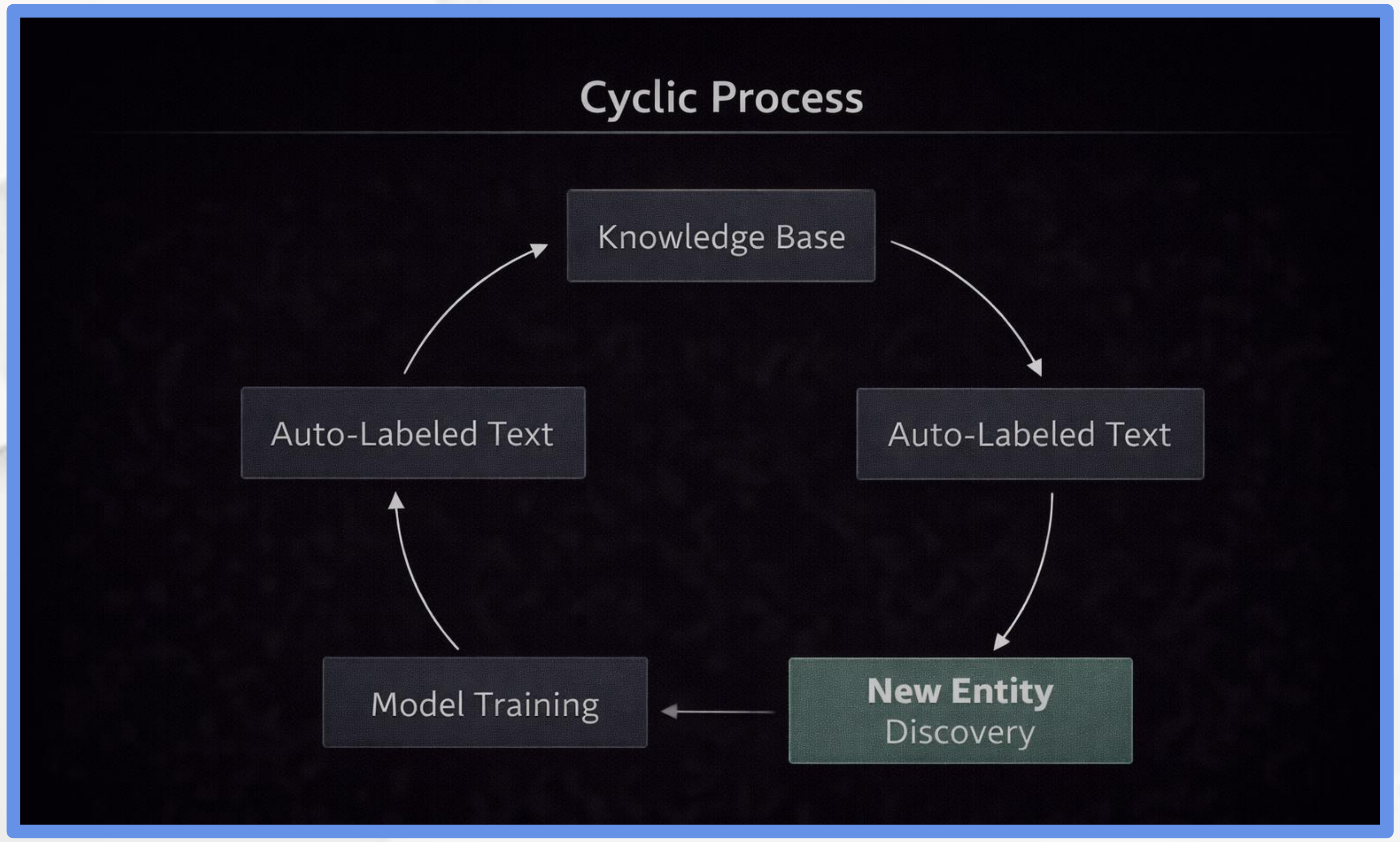
Chapter 4.5 Advanced Domain-Specific Entity Extraction

Distant Supervision and Bootstrapping

- Use existing databases as weak labels.
- Automatically annotate large corpora.
- Iteratively expand entity coverage.
- Produces scalable “silver” datasets.



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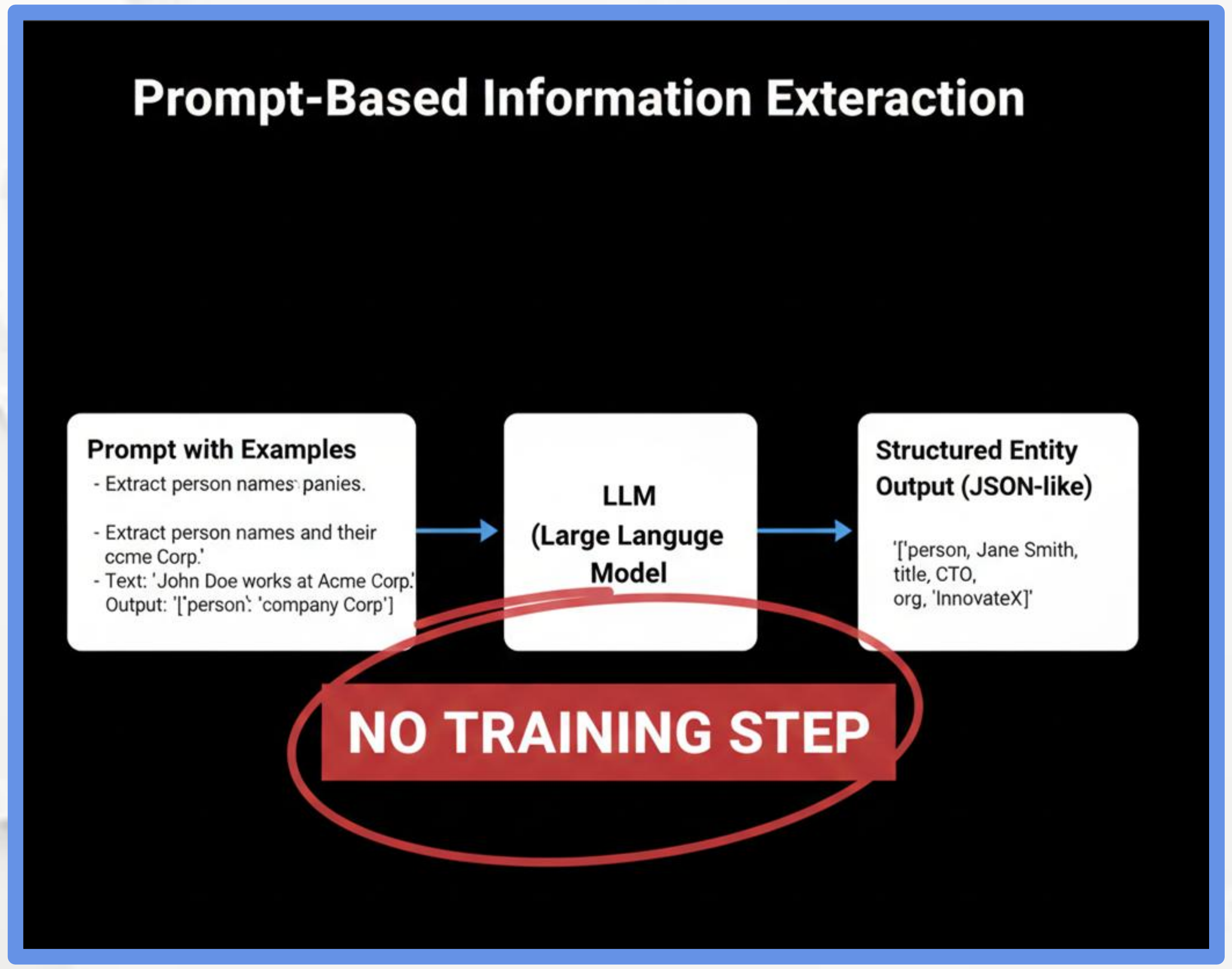
Chapter 4.5 Advanced Domain-Specific Entity Extraction

LLM Prompting for Entity Extraction

- Zero-shot extraction via instructions.
- Few-shot examples improve precision.
- Rapid adaptation without retraining
- Outputs used as silver labels.



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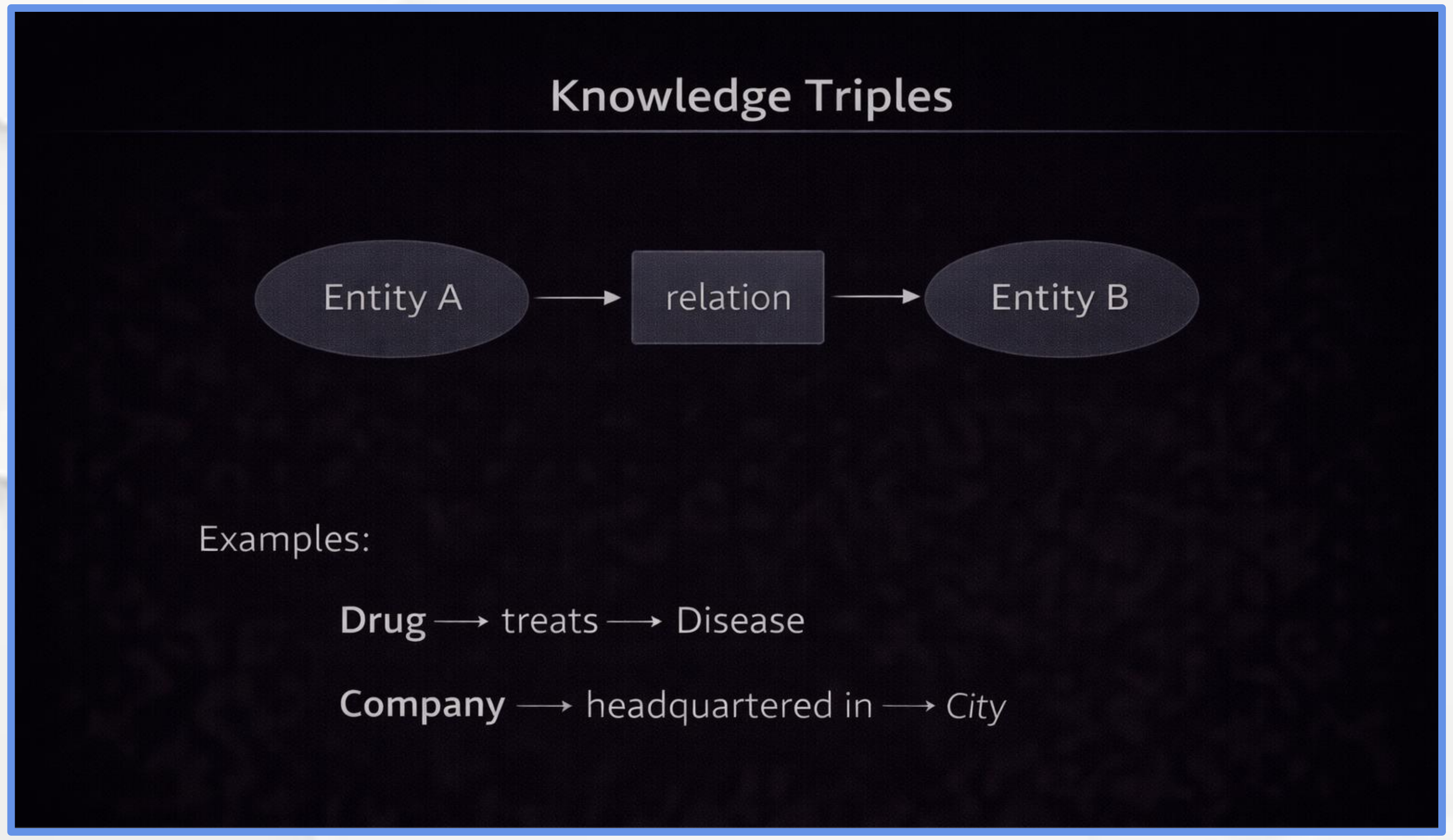
Chapter 4.5 Advanced Domain-Specific Entity Extraction

Relation Extraction as the Next Step

- Entities alone are not sufficient.
- Relations capture semantic structure.
- Output expressed as triples.
- Enables dynamic knowledge graphs.



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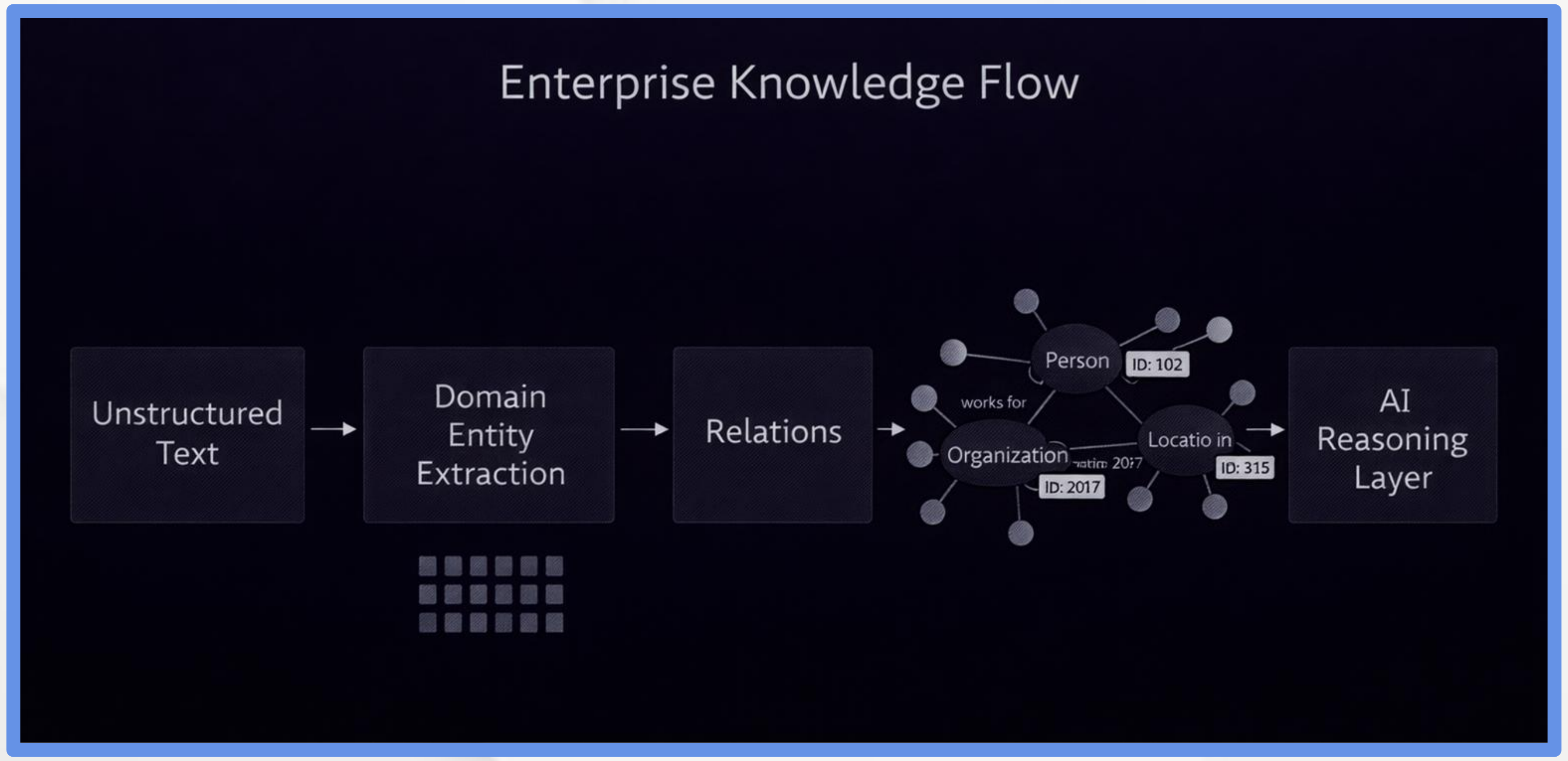
Chapter 4.5 Advanced Domain-Specific Entity Extraction

Why Domain-Specific Extraction Enables Reasoning

- Transforms documents into structured data.
- Supports automated reasoning systems.
- Enables real-time knowledge updates.
- Foundation for decision intelligence.



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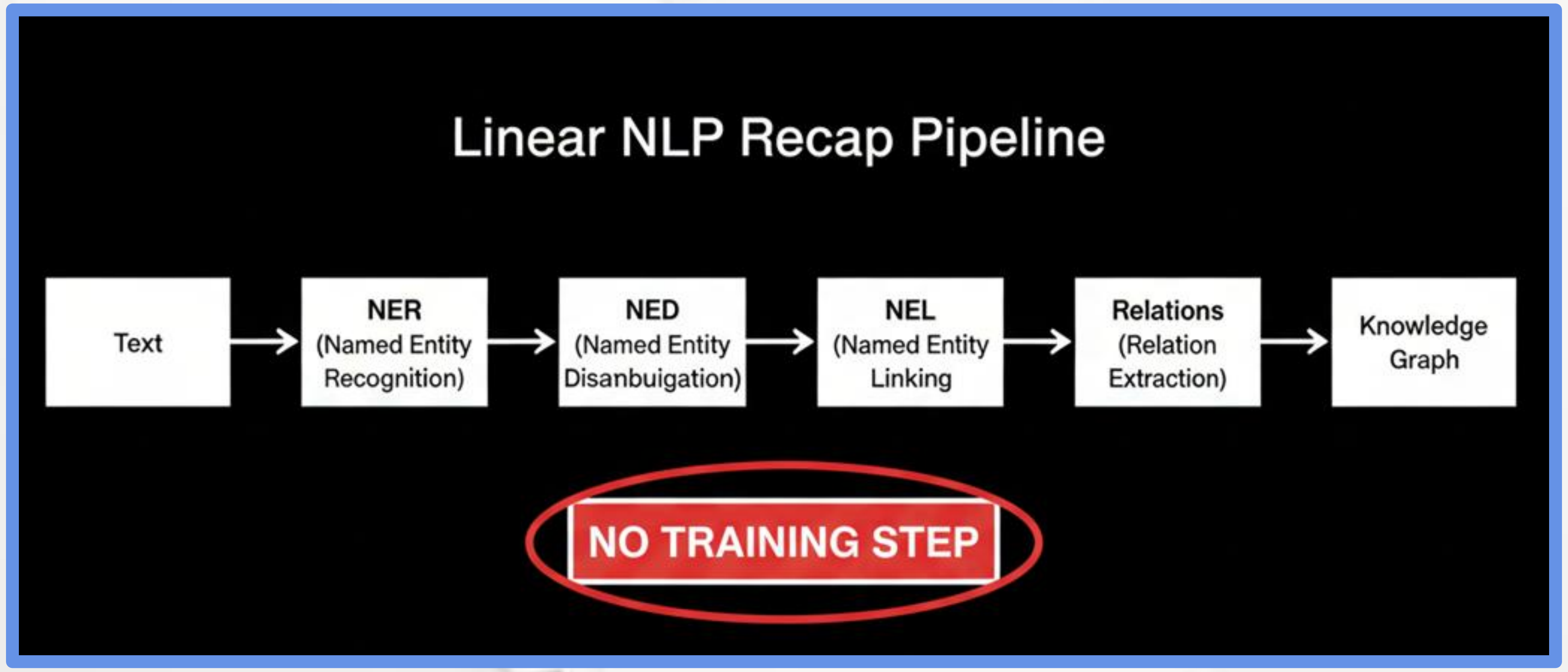
Chapter 4.6 Summary and The Entity Lifecycle

Recap of the Entity-Centric Pipeline

- Entities are the atomic units of meaning.
- NER identifies entity spans.
- NED resolves ambiguity.
- NEL grounds entities in knowledge bases.
- Relations connect entities semantically.



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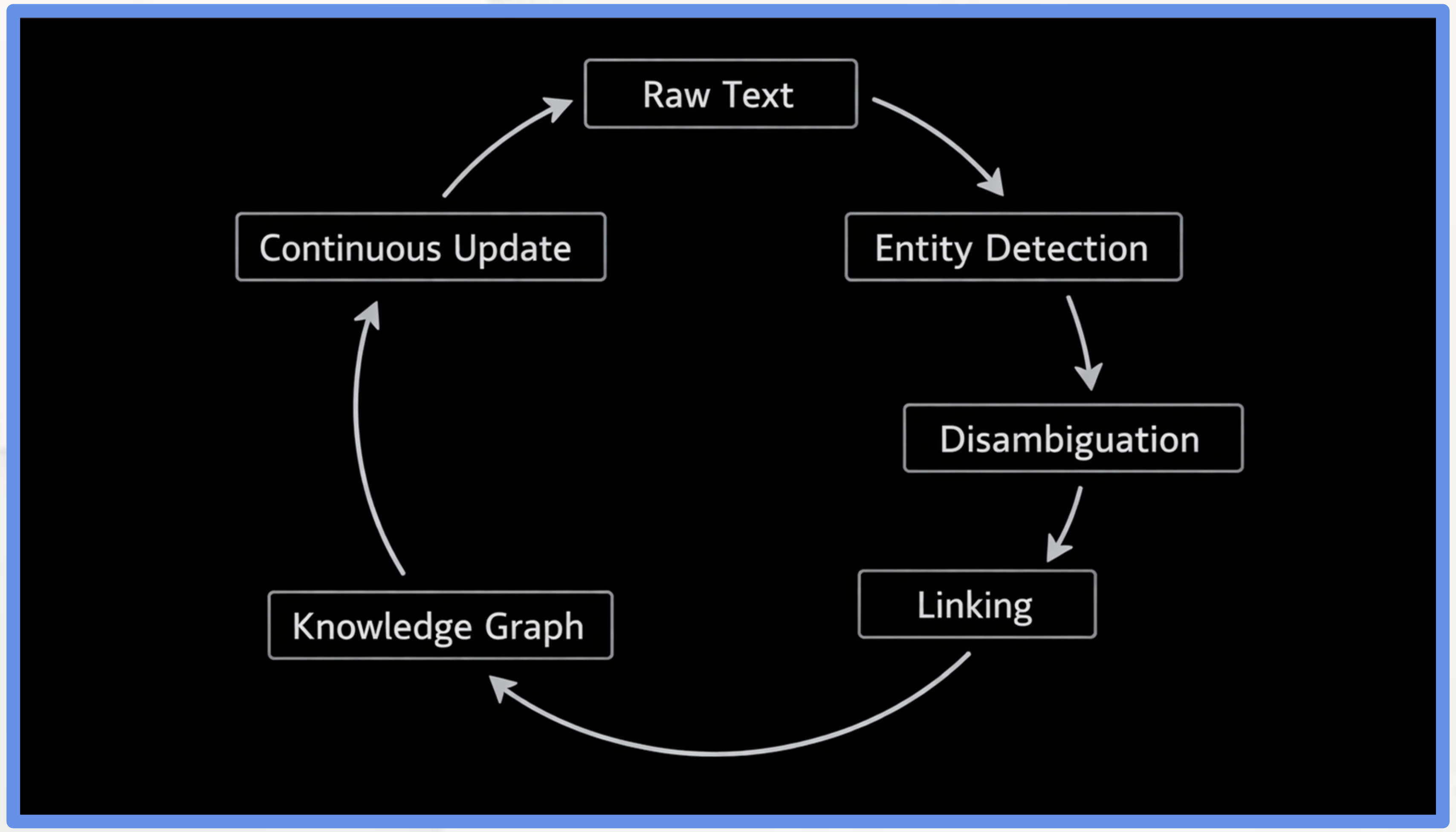
Chapter 4.6 Summary and The Entity Lifecycle

The Complete Entity Lifecycle

- Entities emerge from raw text.
- Entities are validated and linked.
- Entities evolve over time.
- Knowledge graphs are continuously updated.



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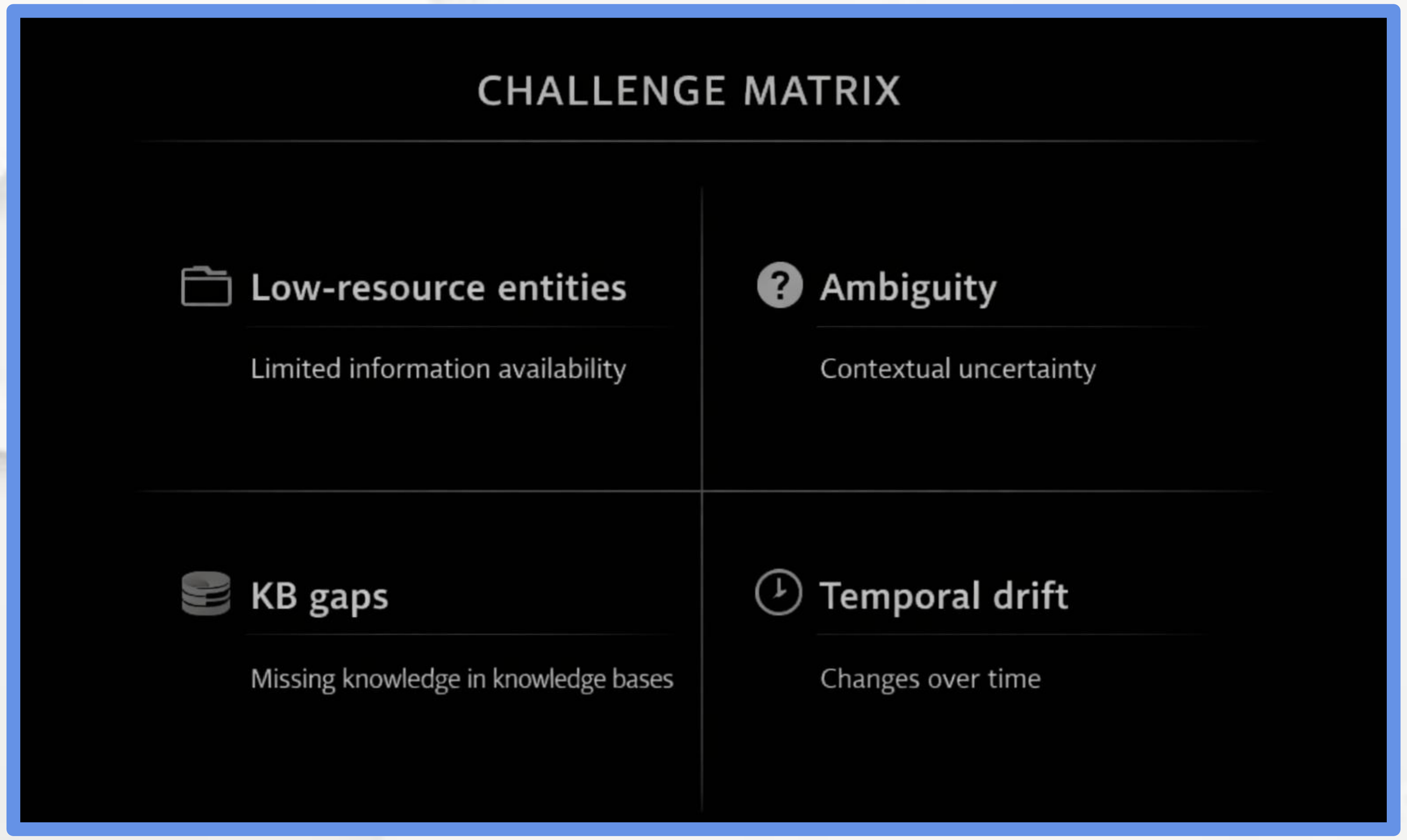
Chapter 4.6 Summary and The Entity Lifecycle

Key Challenges in Real-World Entity Systems

- Low-resource and emerging entities.
- Ambiguous or overlapping mentions.
- Knowledge base incompleteness.
- Entity drift over time.



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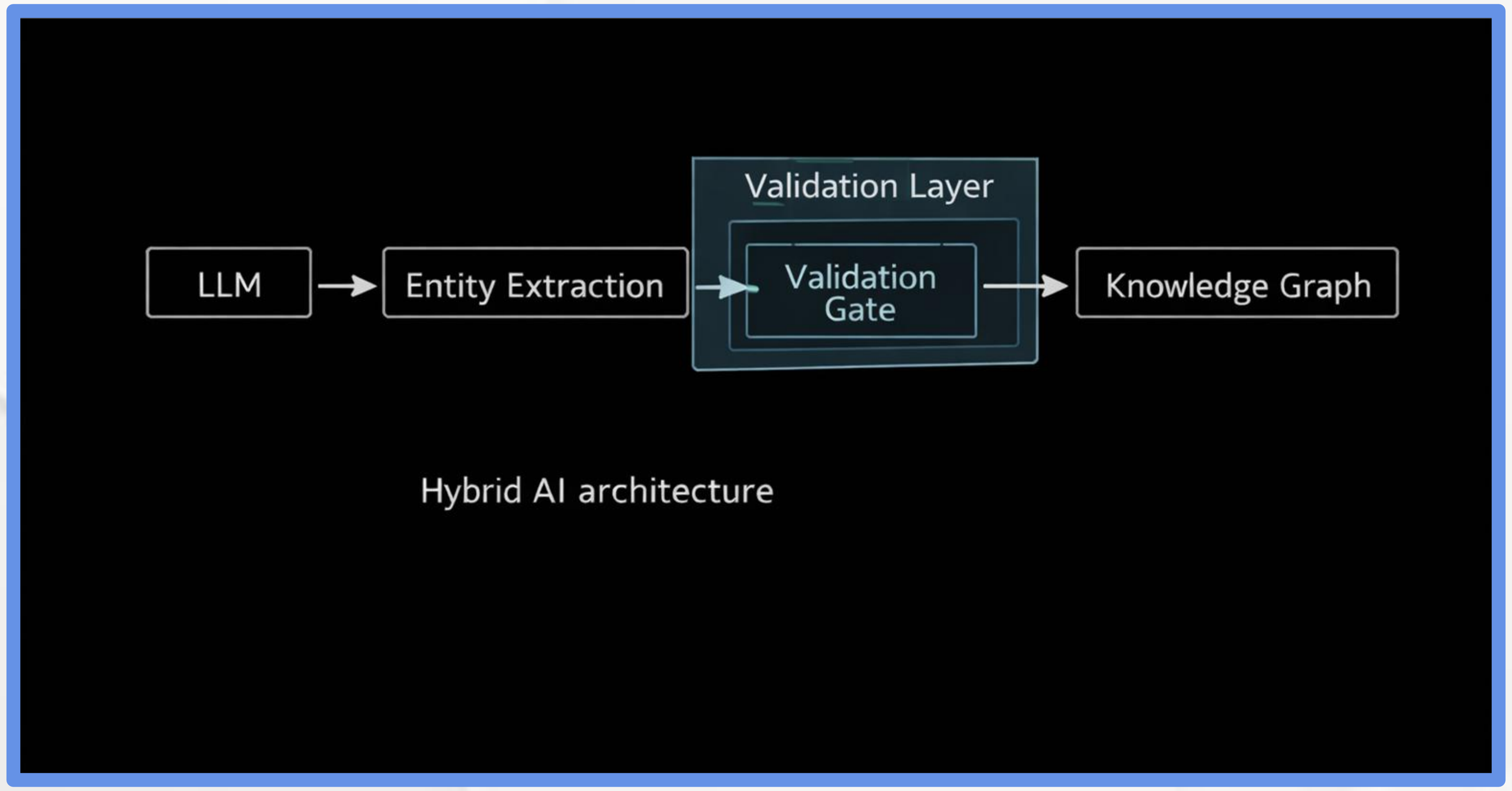
Chapter 4.6 Summary and The Entity Lifecycle

Role of Generative Models in Entity NLP

- LLMs act as internal annotators.
- Enable zero-shot entity extraction.
- Assist relation discovery.
- Must be validated to prevent hallucinations.



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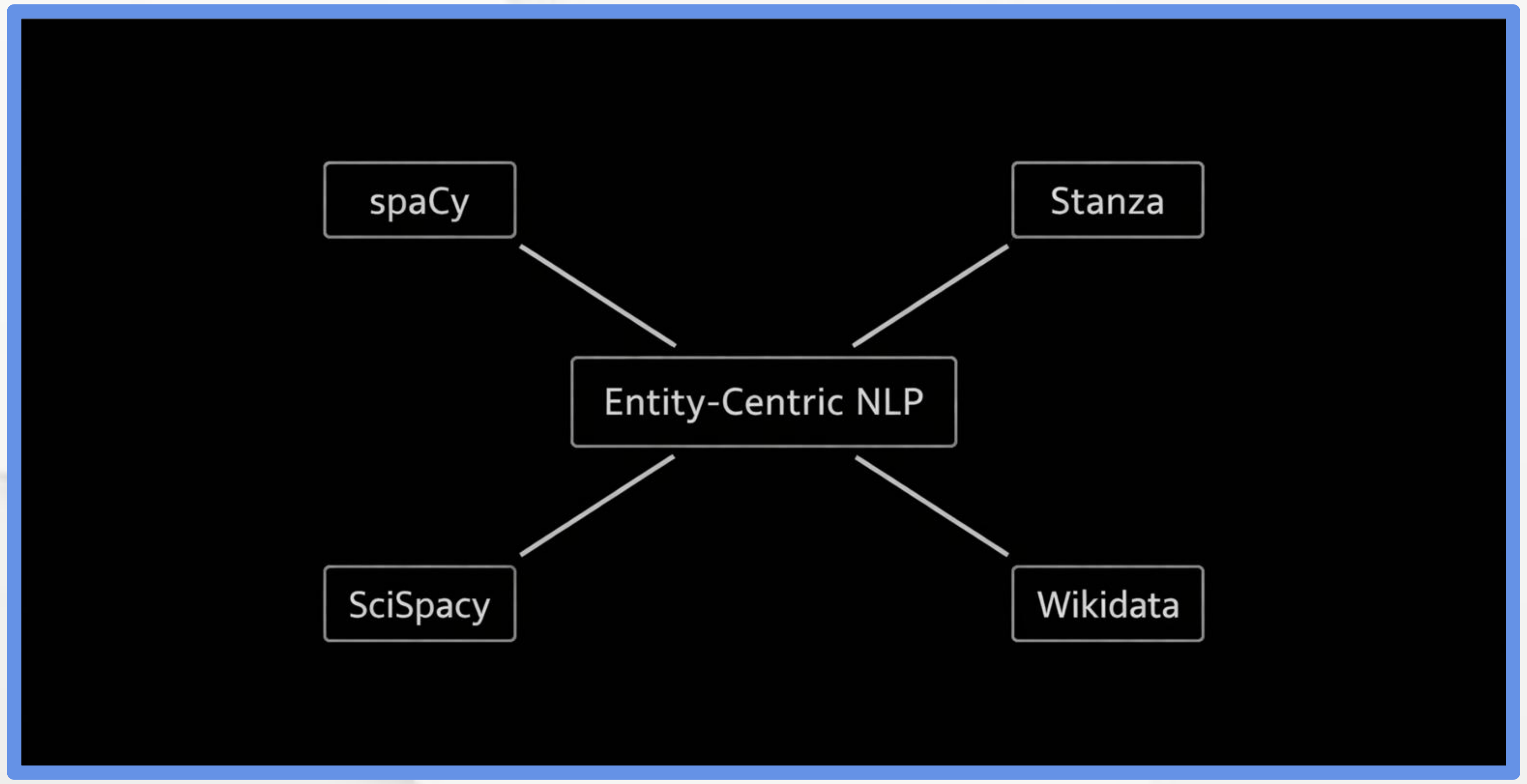
Chapter 4.6 Summary and The Entity Lifecycle

Practical Tools and Libraries

- spaCy: production-grade NER pipelines.
- Stanza: multilingual neural NLP.
- SciSpacy: biomedical entity extraction.
- Wikidata: global knowledge grounding.



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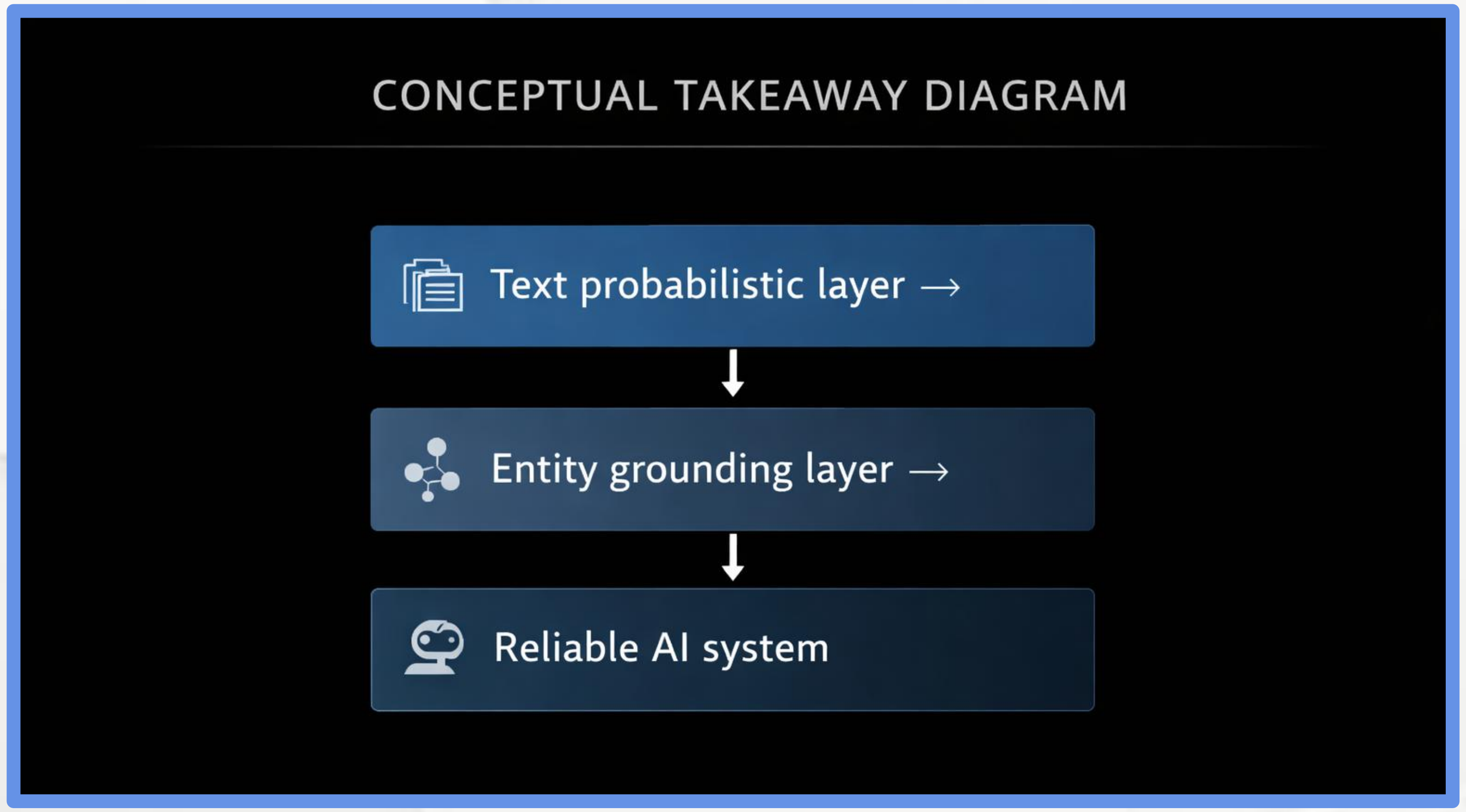
Chapter 4.6 Summary and The Entity Lifecycle

Final Takeaways

- Entities anchor language to reality.
- Entity grounding reduces hallucinations.
- Knowledge graphs enable explainability.
- Entity-centric NLP is essential for trusted AI.



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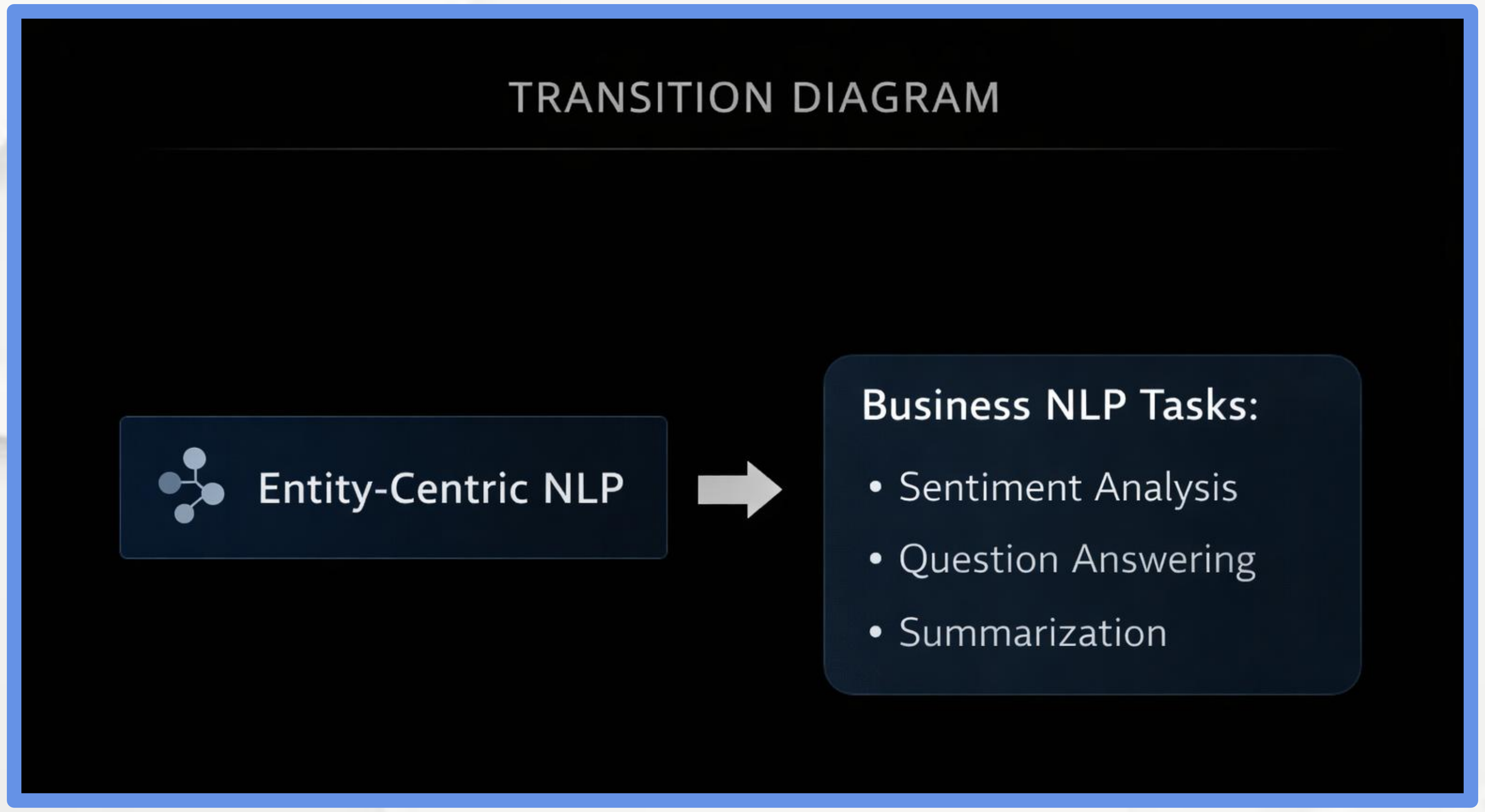
Chapter 4.6 Summary and The Entity Lifecycle

Transition to the Next Module

- Entity-centric NLP enables downstream tasks.
- Grounded entities improve analytics accuracy.
- Next: applying entities to business problems.
- Sentiment, QA, and summarization.



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Chapter 4.6 Summary and The Entity Lifecycle

M04 — Academic References and Research Resources

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